

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397131
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Messerly; Shayne

Automatic dimensional frame reference for fiber optic

Abstract

A system, apparatus and method directed to placing a medical device into a body of a patient, including performing operations of providing a broadband incident light signal to a plurality of core fibers of a multi-core optical fiber, receiving reflected light signals of different spectral width, and processing the reflected light signals associated with the plurality of core fibers to determine (i) a physical state of the multi-core optical fiber relating to the medical device including the multi-core optical fiber, and (ii) an orientation of the multi-core optical fiber relative to a reference frame of the body. Additional operations include generating a display illustrating the physical state of the multi-core optical fiber based at least on the orientation determined during processing of the reflected light. Typically, the display is a two-dimensional representation of the multi-core optical fiber in accordance with the determined orientation.

Inventors:	Messerly; Shayne (Kaysville, UT)
Applicant:	Bard Access Systems, Inc. (Salt Lake City, UT)
Family ID:	1000008781878
Assignee:	Bard Access Systems, Inc. (Salt Lake City, UT)
Appl. No.:	18/538111
Filed:	December 13, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240108856 A1	Apr. 04, 2024

Related U.S. Application Data

continuation parent-doc US 17357186 20210624 US 11883609 child-doc US 18538111
us-provisional-application US 63045667 20200629

Publication Classification

Int. Cl.: **A61B5/06** (20060101); **A61B1/07** (20060101); **A61M25/01** (20060101); A61B34/20 (20160101)

U.S. Cl.:

CPC **A61M25/01** (20130101); **A61B1/07** (20130101); **A61B5/06** (20130101); A61B2034/2055 (20160201)

Field of Classification Search

CPC: A61M (25/01); A61B (1/07); A61B (5/06); A61B (2034/2055)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4813429	12/1988	Eshel et al.	N/A	N/A
5099845	12/1991	Besz et al.	N/A	N/A
5163935	12/1991	Black et al.	N/A	N/A
5207672	12/1992	Roth et al.	N/A	N/A
5211165	12/1992	Dumoulin et al.	N/A	N/A
5275151	12/1993	Shockey et al.	N/A	N/A
5280786	12/1993	Wlodarczyk et al.	N/A	N/A
5423321	12/1994	Fontenot	N/A	N/A
5454807	12/1994	Lennox et al.	N/A	N/A
5460182	12/1994	Goodman et al.	N/A	N/A
5517997	12/1995	Fontenot	N/A	N/A
5622170	12/1996	Schulz	N/A	N/A
5740808	12/1997	Panescu et al.	N/A	N/A
5841131	12/1997	Schroeder et al.	N/A	N/A
5872879	12/1998	Hamm	N/A	N/A
5873842	12/1998	Brennen et al.	N/A	N/A
5879306	12/1998	Fontenot et al.	N/A	N/A
5906579	12/1998	Vander Salm et al.	N/A	N/A
6069698	12/1999	Ozawa et al.	N/A	N/A
6081741	12/1999	Hollis	N/A	N/A
6178346	12/2000	Amundson et al.	N/A	N/A
6208887	12/2000	Clarke	N/A	N/A
6245103	12/2000	Stinson	N/A	N/A
6319227	12/2000	Mansouri-Ruiz	N/A	N/A
6343227	12/2001	Crowley	N/A	N/A
6398721	12/2001	Nakamura et al.	N/A	N/A
6485482	12/2001	Belef	N/A	N/A
6564089	12/2002	Izatt et al.	N/A	N/A
6593884	12/2002	Gilboa et al.	N/A	N/A
6597941	12/2002	Fontenot et al.	N/A	N/A
6650923	12/2002	Lesh et al.	N/A	N/A

6673214	12/2003	Marchitto et al.	N/A	N/A
6685666	12/2003	Fontenot	N/A	N/A
6687010	12/2003	Horii et al.	N/A	N/A
6690966	12/2003	Rava et al.	N/A	N/A
6701181	12/2003	Tang et al.	N/A	N/A
6711426	12/2003	Benaron et al.	N/A	N/A
6816743	12/2003	Moreno et al.	N/A	N/A
6892090	12/2004	Verard et al.	N/A	N/A
6895267	12/2004	Panescu et al.	N/A	N/A
7132645	12/2005	Korn	N/A	N/A
7273056	12/2006	Wilson et al.	N/A	N/A
7344533	12/2007	Pearson et al.	N/A	N/A
7366562	12/2007	Dukesherer et al.	N/A	N/A
7366563	12/2007	Kleen et al.	N/A	N/A
7396354	12/2007	Rychnovsky et al.	N/A	N/A
7406346	12/2007	Kleen et al.	N/A	N/A
7515265	12/2008	Alfano et al.	N/A	N/A
7532920	12/2008	Ainsworth et al.	N/A	N/A
7587236	12/2008	Demos et al.	N/A	N/A
7603166	12/2008	Casscells, III et al.	N/A	N/A
7729735	12/2009	Burchman	N/A	N/A
7757695	12/2009	Wilson et al.	N/A	N/A
7758499	12/2009	Adler	N/A	N/A
7840253	12/2009	Tremblay et al.	N/A	N/A
7992573	12/2010	Wilson et al.	N/A	N/A
8032200	12/2010	Tearney et al.	N/A	N/A
8054469	12/2010	Nakabayashi et al.	N/A	N/A
8060187	12/2010	Marshik-Geurts et al.	N/A	N/A
8073517	12/2010	Burchman	N/A	N/A
8078261	12/2010	Imam	N/A	N/A
8187189	12/2011	Jung et al.	N/A	N/A
8267932	12/2011	Baxter et al.	N/A	N/A
8369932	12/2012	Cinbis et al.	N/A	N/A
8388541	12/2012	Messerly et al.	N/A	N/A
8571640	12/2012	Holman	N/A	N/A
8597315	12/2012	Snow et al.	N/A	N/A
8700358	12/2013	Parker, Jr.	N/A	N/A
8781555	12/2013	Burnside et al.	N/A	N/A
8798721	12/2013	Dib	N/A	N/A
8968331	12/2014	Sochor	N/A	N/A
8979871	12/2014	Tyc et al.	N/A	N/A
9060687	12/2014	Yamanaka et al.	N/A	N/A
9114226	12/2014	Lash et al.	N/A	N/A
9206309	12/2014	Appleby et al.	N/A	N/A
9360630	12/2015	Jenner et al.	N/A	N/A
9504392	12/2015	Caron et al.	N/A	N/A
9560954	12/2016	Jacobs et al.	N/A	N/A
9622706	12/2016	Dick et al.	N/A	N/A
9678275	12/2016	Griffin	N/A	N/A

10231753	12/2018	Burnside et al.	N/A	N/A
10258240	12/2018	Eberle et al.	N/A	N/A
10327830	12/2018	Grant et al.	N/A	N/A
10349890	12/2018	Misener et al.	N/A	N/A
10492876	12/2018	Anastassiou et al.	N/A	N/A
10568586	12/2019	Begin et al.	N/A	N/A
10631718	12/2019	Petroff et al.	N/A	N/A
10992078	12/2020	Thompson et al.	N/A	N/A
11123047	12/2020	Jaffer et al.	N/A	N/A
11525670	12/2021	Messerly et al.	N/A	N/A
2001/0014793	12/2000	Brugger et al.	N/A	N/A
2002/0198457	12/2001	Tearney et al.	N/A	N/A
2003/0092995	12/2002	Thompson	N/A	N/A
2004/0129555	12/2003	Marchitto et al.	N/A	N/A
2004/0161362	12/2003	Bogert	N/A	N/A
2004/0242995	12/2003	Maschke	N/A	N/A
2005/0033264	12/2004	Redinger	N/A	N/A
2005/0163424	12/2004	Chen	N/A	N/A
2005/0261598	12/2004	Banet et al.	N/A	N/A
2005/0278010	12/2004	Richardson	N/A	N/A
2006/0013523	12/2005	Childlers et al.	N/A	N/A
2006/0015136	12/2005	Besselink	N/A	N/A
2006/0036164	12/2005	Wilson et al.	N/A	N/A
2006/0189959	12/2005	Schneider	N/A	N/A
2006/0200049	12/2005	Leo et al.	N/A	N/A
2006/0241395	12/2005	Kruger et al.	N/A	N/A
2006/0241492	12/2005	Boese et al.	N/A	N/A
2007/0156019	12/2006	Larkin et al.	N/A	N/A
2007/0201793	12/2006	Askins et al.	N/A	N/A
2007/0287886	12/2006	Saadat	N/A	N/A
2007/0287934	12/2006	Babaev	N/A	N/A
2007/0299425	12/2006	Waner et al.	N/A	N/A
2008/0039715	12/2007	Wilson et al.	N/A	N/A
2008/0077225	12/2007	Carlin et al.	N/A	N/A
2008/0082004	12/2007	Banet et al.	N/A	N/A
2008/0172119	12/2007	Yamasaki et al.	N/A	N/A
2008/0183128	12/2007	Morriss et al.	N/A	N/A
2008/0212082	12/2007	Froggatt et al.	N/A	N/A
2008/0285909	12/2007	Younge et al.	N/A	N/A
2008/0319290	12/2007	Mao et al.	N/A	N/A
2009/0054908	12/2008	Zand et al.	N/A	N/A
2009/0062634	12/2008	Say et al.	N/A	N/A
2009/0118612	12/2008	Grunwald et al.	N/A	N/A
2009/0137952	12/2008	Ramamurthy et al.	N/A	N/A
2009/0234328	12/2008	Cox et al.	N/A	N/A
2009/0304582	12/2008	Rousso et al.	N/A	N/A
2009/0314925	12/2008	Van Vorhis et al.	N/A	N/A
2010/0016729	12/2009	Futrell	N/A	N/A
2010/0030063	12/2009	Lee et al.	N/A	N/A
2010/0030132	12/2009	Niezgoda et al.	N/A	N/A

2010/0063534	12/2009	Kugler et al.	N/A	N/A
2010/0114115	12/2009	Schlesinger et al.	N/A	N/A
2010/0286531	12/2009	Ryan et al.	N/A	N/A
2010/0312095	12/2009	Jenkins et al.	N/A	N/A
2011/0060229	12/2010	Hulvershorn et al.	N/A	N/A
2011/0090486	12/2010	Udd	N/A	N/A
2011/0144481	12/2010	Feer et al.	N/A	N/A
2011/0166442	12/2010	Sarvazyan	N/A	N/A
2011/0172591	12/2010	Babaev	N/A	N/A
2011/0172680	12/2010	Younge et al.	N/A	N/A
2011/0237958	12/2010	Onimura	N/A	N/A
2011/0242532	12/2010	McKenna	N/A	N/A
2011/0245662	12/2010	Eggers et al.	N/A	N/A
2011/0288405	12/2010	Razavi et al.	N/A	N/A
2011/0295108	12/2010	Cox et al.	N/A	N/A
2011/0313280	12/2010	Govari et al.	N/A	N/A
2012/0046562	12/2011	Powers et al.	N/A	N/A
2012/0065481	12/2011	Hunter et al.	N/A	N/A
2012/0101413	12/2011	Beetel et al.	N/A	N/A
2012/0136242	12/2011	Qi et al.	N/A	N/A
2012/0143029	12/2011	Silverstein et al.	N/A	N/A
2012/0184827	12/2011	Shwartz et al.	N/A	N/A
2012/0184955	12/2011	Pivotto et al.	N/A	N/A
2012/0283747	12/2011	Popovic	N/A	N/A
2012/0289783	12/2011	Duindam et al.	N/A	N/A
2012/0321243	12/2011	Younge et al.	N/A	N/A
2013/0028554	12/2012	Wong et al.	N/A	N/A
2013/0072943	12/2012	Parmar	N/A	N/A
2013/0096482	12/2012	Bertrand et al.	N/A	N/A
2013/0104884	12/2012	Vazales et al.	N/A	N/A
2013/0188855	12/2012	Desjardins et al.	N/A	N/A
2013/0204124	12/2012	Duindam et al.	N/A	N/A
2013/0211246	12/2012	Parasher	N/A	N/A
2013/0296693	12/2012	Wenzel et al.	N/A	N/A
2013/0310668	12/2012	Young	N/A	N/A
2013/0317372	12/2012	Eberle et al.	N/A	N/A
2013/0324840	12/2012	Zhongping et al.	N/A	N/A
2014/0121468	12/2013	Eichenholz	N/A	N/A
2014/0221829	12/2013	Maitland et al.	N/A	N/A
2014/0275997	12/2013	Chopra et al.	N/A	N/A
2015/0029511	12/2014	'T Hoof et al.	N/A	N/A
2015/0031987	12/2014	Pameijer et al.	N/A	N/A
2015/0080688	12/2014	Cinbis et al.	N/A	N/A
2015/0099979	12/2014	Caves et al.	N/A	N/A
2015/0119700	12/2014	Liang et al.	N/A	N/A
2015/0124264	12/2014	Ramachandran et al.	N/A	N/A
2015/0141854	12/2014	Eberle et al.	N/A	N/A
2015/0164571	12/2014	Saadat	N/A	N/A
2015/0190221	12/2014	Schaefer et al.	N/A	N/A
2015/0209113	12/2014	Burkholz et al.	N/A	N/A

2015/0209117	12/2014	Flexman et al.	N/A	N/A
2015/0254526	12/2014	Denissen	N/A	N/A
2015/0320977	12/2014	Vitullo et al.	N/A	N/A
2016/0018602	12/2015	Govari et al.	N/A	N/A
2016/0067449	12/2015	Misener et al.	N/A	N/A
2016/0102969	12/2015	Verstege et al.	N/A	N/A
2016/0166326	12/2015	Bakker et al.	N/A	N/A
2016/0166341	12/2015	Iordachita et al.	N/A	N/A
2016/0184020	12/2015	Kowalewski et al.	N/A	N/A
2016/0213432	12/2015	Flexman et al.	N/A	N/A
2016/0331461	12/2015	Cheatham, III et al.	N/A	N/A
2016/0349044	12/2015	Marell et al.	N/A	N/A
2016/0354038	12/2015	Demirtas et al.	N/A	N/A
2017/0020394	12/2016	Harrington	N/A	N/A
2017/0079681	12/2016	Burnside et al.	N/A	N/A
2017/0082806	12/2016	Van Der Mark et al.	N/A	N/A
2017/0151027	12/2016	Walker et al.	N/A	N/A
2017/0173349	12/2016	Pfleiderer et al.	N/A	N/A
2017/0196479	12/2016	Liu et al.	N/A	N/A
2017/0201036	12/2016	Cohen et al.	N/A	N/A
2017/0215973	12/2016	Flexman et al.	N/A	N/A
2017/0231699	12/2016	Flexman et al.	N/A	N/A
2017/0273542	12/2016	Au	N/A	N/A
2017/0273565	12/2016	Ma et al.	N/A	N/A
2017/0273628	12/2016	Ofek et al.	N/A	N/A
2017/0290563	12/2016	Cole et al.	N/A	N/A
2017/0311901	12/2016	Zhao et al.	N/A	N/A
2017/0319279	12/2016	Fish et al.	N/A	N/A
2018/0008443	12/2017	Cole et al.	N/A	N/A
2018/0031493	12/2017	Tojo et al.	N/A	N/A
2018/0095231	12/2017	Lowell et al.	N/A	N/A
2018/0113038	12/2017	Janabi-Sharifi et al.	N/A	N/A
2018/0116551	12/2017	Newman et al.	N/A	N/A
2018/0140170	12/2017	Van Putten et al.	N/A	N/A
2018/0235709	12/2017	Donhowe et al.	N/A	N/A
2018/0239124	12/2017	Naruse et al.	N/A	N/A
2018/0240237	12/2017	Donhowe et al.	N/A	N/A
2018/0250088	12/2017	Brennan et al.	N/A	N/A
2018/0264227	12/2017	Flexman et al.	N/A	N/A
2018/0279909	12/2017	Noonan et al.	N/A	N/A
2018/0289390	12/2017	Amorizzo et al.	N/A	N/A
2018/0289927	12/2017	Messerly	N/A	N/A
2018/0339134	12/2017	Leo	N/A	N/A
2018/0360545	12/2017	Cole et al.	N/A	N/A
2019/0059743	12/2018	Ramachandran et al.	N/A	N/A
2019/0060003	12/2018	Tuason	N/A	N/A
2019/0110844	12/2018	Misener et al.	N/A	N/A
2019/0231272	12/2018	Yamaji	N/A	N/A
2019/0237902	12/2018	Thompson et al.	N/A	N/A
2019/0247132	12/2018	Harks et al.	N/A	N/A

2019/0307331	12/2018	Saadat et al.	N/A	N/A
2019/0321110	12/2018	Grunwald et al.	N/A	N/A
2019/0343424	12/2018	Blumenkranz et al.	N/A	N/A
2019/0343702	12/2018	Smith	N/A	N/A
2019/0357875	12/2018	Qi et al.	N/A	N/A
2019/0365199	12/2018	Zhao et al.	N/A	N/A
2019/0374130	12/2018	Bydlon et al.	N/A	N/A
2020/0000526	12/2019	Zhao	N/A	N/A
2020/0030575	12/2019	Bogusky et al.	N/A	N/A
2020/0046434	12/2019	Graetzel et al.	N/A	N/A
2020/0054399	12/2019	Duindam et al.	N/A	N/A
2020/0121482	12/2019	Spector et al.	N/A	N/A
2020/0188036	12/2019	Ding et al.	N/A	N/A
2020/0297442	12/2019	Adebar et al.	N/A	N/A
2020/0305983	12/2019	Yampolsky et al.	N/A	N/A
2020/0315770	12/2019	Dupont et al.	N/A	N/A
2020/0394789	12/2019	Freund et al.	N/A	N/A
2021/0015470	12/2020	Prisco et al.	N/A	N/A
2021/0023341	12/2020	Decheek et al.	N/A	N/A
2021/0045814	12/2020	Thompson et al.	N/A	N/A
2021/0068911	12/2020	Walker et al.	N/A	N/A
2021/0100627	12/2020	Soper	N/A	G06T 7/0014
2021/0244311	12/2020	Zhao et al.	N/A	N/A
2021/0268229	12/2020	Sowards et al.	N/A	N/A
2021/0271035	12/2020	Sowards et al.	N/A	N/A
2021/0275257	12/2020	Prior et al.	N/A	N/A
2021/0298680	12/2020	Sowards et al.	N/A	N/A
2021/0330399	12/2020	Netravali et al.	N/A	N/A
2021/0401456	12/2020	Cox et al.	N/A	N/A
2021/0401509	12/2020	Misener et al.	N/A	N/A
2021/0402144	12/2020	Messerly	N/A	N/A
2022/0011192	12/2021	Misener et al.	N/A	N/A
2022/0034733	12/2021	Misener et al.	N/A	N/A
2022/0079683	12/2021	Bydlon et al.	N/A	N/A
2022/0096796	12/2021	McLaughlin et al.	N/A	N/A
2022/0110508	12/2021	Van Roosbroeck et al.	N/A	N/A
2022/0110695	12/2021	Sowards et al.	N/A	N/A
2022/0151568	12/2021	Yao et al.	N/A	N/A
2022/0152349	12/2021	Sowards et al.	N/A	N/A
2022/0160209	12/2021	Sowards et al.	N/A	N/A
2022/0172354	12/2021	Misener et al.	N/A	N/A
2022/0211442	12/2021	McLaughlin et al.	N/A	N/A
2022/0233246	12/2021	Misener et al.	N/A	N/A
2022/0369934	12/2021	Sowards et al.	N/A	N/A
2023/0081198	12/2022	Sowards et al.	N/A	N/A
2023/0097431	12/2022	Sowards et al.	N/A	N/A
2023/0101030	12/2022	Misener et al.	N/A	N/A
2023/0108604	12/2022	Messerly et al.	N/A	N/A

2023/0126813	12/2022	Sowards et al.	N/A	N/A
2023/0243715	12/2022	Misener et al.	N/A	N/A
2023/0248444	12/2022	Misener et al.	N/A	N/A
2023/0251150	12/2022	Misener et al.	N/A	N/A
2023/0337985	12/2022	Sowards et al.	N/A	N/A
2023/0346479	12/2022	Muller et al.	N/A	N/A
2023/0414112	12/2022	Misener et al.	N/A	N/A
2024/0000515	12/2023	Misener et al.	N/A	N/A
2024/0050708	12/2023	Misener	N/A	N/A
2024/0099659	12/2023	Sowards et al.	N/A	N/A
2024/0216077	12/2023	Thompson et al.	N/A	N/A
2024/0335237	12/2023	Sowards et al.	N/A	N/A
2024/0353275	12/2023	Misener et al.	N/A	N/A
2024/0383133	12/2023	Bydlon et al.	N/A	N/A
2024/0390644	12/2023	McLaughlin et al.	N/A	N/A
2025/0020453	12/2024	Thompson et al.	N/A	N/A
2025/0060266	12/2024	Messerly	N/A	N/A
2025/0186135	12/2024	McLaughlin et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101132730	12/2007	CN	N/A
111265309	12/2019	CN	N/A
113080937	12/2020	CN	N/A
102016109601	12/2016	DE	N/A
2240111	12/2009	EP	N/A
2907445	12/2014	EP	N/A
3545849	12/2018	EP	N/A
3705020	12/2019	EP	N/A
7366562	12/2022	JP	N/A
20190098512	12/2018	KR	N/A
9964099	12/1998	WO	N/A
2006113394	12/2005	WO	N/A
2006122001	12/2005	WO	N/A
2007002323	12/2006	WO	N/A
2009155325	12/2008	WO	N/A
2011121516	12/2010	WO	N/A
2011141830	12/2010	WO	N/A
2011150376	12/2010	WO	N/A
2012064769	12/2011	WO	N/A
2015044930	12/2014	WO	N/A
2015074045	12/2014	WO	N/A
2016038492	12/2015	WO	N/A
2016051302	12/2015	WO	N/A
2016061431	12/2015	WO	N/A
2016149819	12/2015	WO	N/A
2018096491	12/2017	WO	N/A
2019037071	12/2018	WO	N/A
2019046769	12/2018	WO	N/A
2019070423	12/2018	WO	N/A

2019230713	12/2018	WO	N/A
2020182997	12/2019	WO	N/A
2021030092	12/2020	WO	N/A
2021108688	12/2020	WO	N/A
2021108697	12/2020	WO	N/A
2021138096	12/2020	WO	N/A
2021216089	12/2020	WO	N/A
2022031613	12/2021	WO	N/A
2022081723	12/2021	WO	N/A
2022150411	12/2021	WO	N/A
2022164902	12/2021	WO	N/A
2022245987	12/2021	WO	N/A
2023043954	12/2022	WO	N/A
2023049443	12/2022	WO	N/A
2023055810	12/2022	WO	N/A
2023076143	12/2022	WO	N/A
2023211752	12/2022	WO	N/A
2024006384	12/2023	WO	N/A
2024006441	12/2023	WO	N/A
2024064334	12/2023	WO	N/A
2024215665	12/2023	WO	N/A

OTHER PUBLICATIONS

PCT/US2023/033471 filed Sep. 22, 2023 International Search Report and Written Opinion dated Dec. 21, 2023. cited by applicant

U.S. Appl. No. 17/500,678, filed Oct. 13, 2021 Non-Final Office Action dated Jan. 19, 2024. cited by applicant

U.S. Appl. No. 18/079,653, filed Dec. 12, 2022 Non-Final Office Action dated Feb. 6, 2024. cited by applicant

U.S. Appl. No. 18/135,337, filed Apr. 17, 2023 Notice of Allowance dated Mar. 8, 2024. cited by applicant

U.S. Appl. No. 17/500,678, filed Oct. 13, 2021 Final Office Action dated Sep. 21, 2023. cited by applicant

U.S. Appl. No. 18/135,337, filed Apr. 17, 2023 Non-Final Office Action dated Dec. 22, 2023. cited by applicant

U.S. Appl. No. 15/947,267, filed Apr. 6, 2018 Final Office Action dated Jun. 30, 2021. cited by applicant

U.S. Appl. No. 15/947,267, filed Apr. 6, 2018 Final Office Action dated Nov. 10, 2020. cited by applicant

U.S. Appl. No. 15/947,267, filed Apr. 6, 2018 Non-Final Office Action dated Mar. 12, 2021. cited by applicant

U.S. Appl. No. 15/947,267, filed Apr. 6, 2018 Non-Final Office Action dated May 29, 2020. cited by applicant

U.S. Appl. No. 15/947,267, filed Apr. 6, 2018 Non-Final Office Action dated Oct. 13, 2021. cited by applicant

U.S. Appl. No. 17/500,678, filed Oct. 13, 2021 Non-Final Office Action dated Mar. 15, 2023. cited by applicant

U.S. Appl. No. 17/569,350, filed Jan. 5, 2022 Non-Final Office Action dated Jan. 8, 2024. cited by applicant

Dziuda L et al: "Monitoring Respiration and Cardiac Activity Using Fiber Bragg Grating-Based

Sensor”, IEEE Transactions On Biomedical Engineering vol. 59, No. 7, Jul. 2012 pp. 1934-1942. cited by applicant

Dziuda L. et al: “Fiber-optic sensor for monitoring respiration and cardiac activity”, 2011 IEEE Sensors Proceedings : Limerick, Ireland, Oct. 2011 pp. 413-416. cited by applicant

EP 20853352.1 filed Mar. 7, 2022 Extended European Search Report dated Jul. 27, 2023. cited by applicant

EP 20893677.3 filed Jun. 22, 2022 Extended European Search Report dated Oct. 13, 2023. cited by applicant

EP 20894633.5 filed Jun. 22, 2022 Extended European Search Report dated Oct. 16, 2023. cited by applicant

Fiber Optic RealShape (FORS) technology—research. Philips. (Oct. 18, 2018). Retrieved Feb. 28, 2023, from [https:// www.philips.com/a-w/research/research-programs/fors.html](https://www.philips.com/a-w/research/research-programs/fors.html) (Year: 2018). cited by applicant

Jackle Sonja et al. “Three-dimensional guidance including shape sensing of a stentgraft system for endovascular aneurysm repair.” International Journal of Computer Assisted Radiology and Surgery, Springer DE. vol. 15, No. 6, May 7, 2020. cited by applicant

PCT/US2018/026493 filed Apr. 6, 2018 International Search Report and Written Opinion dated Jun. 22, 2018. cited by applicant

PCT/US2020/044801 filed Aug. 3, 2020 International Search Report and Written Opinion dated Oct. 26, 2020. cited by applicant

PCT/US2020/062396 filed Nov. 25, 2020 International Preliminary Report on Patentability dated Jan. 29, 2021. cited by applicant

PCT/US2020/062396 filed Nov. 25, 2020 International Search Report and Written Opinion dated Mar. 2, 2021. cited by applicant

PCT/US2020/062407 filed Nov. 25, 2020 International Preliminary Report on Patentability dated Jan. 25, 2021. cited by applicant

PCT/US2020/062407 filed Nov. 25, 2020 International Search Report and Written Opinion dated Mar. 11, 2021. cited by applicant

PCT/US2021/038899 filed Jun. 24, 2021 International Search Report and Written Opinion dated Oct. 6, 2021. cited by applicant

PCT/US2021/038954 filed Jun. 24, 2021 International Search Report and Written Opinion dated Oct. 28, 2021. cited by applicant

PCT/US2021/041128 filed Jul. 9, 2021 International Search Report and Written Opinion dated Oct. 25, 2021. cited by applicant

PCT/US2021/044216 filed Aug. 2, 2021 International Search Report and Written Opinion dated Nov. 18, 2021. cited by applicant

PCT/US2021/052046 filed Sep. 24, 2021 International Search Report and Written Opinion dated Jan. 11, 2022. cited by applicant

PCT/US2021/054802 filed Oct. 13, 2021 International Search Report and Written Opinion dated Feb. 2, 2022. cited by applicant

PCT/US2022/011347 filed Jan. 5, 2022 International Search Report and Written Opinion dated May 3, 2022. cited by applicant

PCT/US2022/013897 filed Jan. 26, 2022 International Search Report and Written Opinion dated May 11, 2022. cited by applicant

PCT/US2022/029894 filed May 18, 2022, International Search Report and Written Opinion dated Sep. 1, 2022. cited by applicant

PCT/US2022/043706 filed Sep. 16, 2022 International Search Report and Written Opinion dated Nov. 24, 2022. cited by applicant

PCT/US2022/044696 filed Sep. 26, 2022 International Search Report and Written Opinion dated Jan. 23, 2023. cited by applicant

PCT/US2022/045051 filed Sep. 28, 2022 International Search Report and Written Opinion dated Jan. 2, 2023. cited by applicant
PCT/US2022/047538 filed Oct. 24, 2022 International Search Report and Written Opinion dated Jan. 26, 2023. cited by applicant
PCT/US2023/019239 filed Apr. 20, 2023 International Search Report and Written Opinion dated Jul. 20, 2023. cited by applicant
PCT/US2023/026487 filed Jun. 28, 2023 International Search Report and Written Opinion dated Sep. 6, 2023. cited by applicant
PCT/US2023/026581 filed Jun. 29, 2023 International Search Report and Written Opinion dated Oct. 27, 2023. cited by applicant
U.S. Appl. No. 15/947,267, filed Apr. 6, 2018 Examiner's Answer dated Nov. 28, 2022. cited by applicant
U.S. Appl. No. 15/947,267, filed Apr. 6, 2018 Final Office Action dated Apr. 22, 2022. cited by applicant
U.S. Appl. No. 16/984,104, filed Aug. 3, 2020 Final Office Action dated Sep. 21, 2023. cited by applicant
U.S. Appl. No. 16/984,104, filed Aug. 3, 2020 Non-Final Office Action dated Jun. 22, 2023. cited by applicant
U.S. Appl. No. 16/984,104, filed Aug. 3, 2020 Notice of Allowance dated Nov. 21, 2023. cited by applicant
U.S. Appl. No. 16/984,104, filed Aug. 3, 2020 Restriction Requirement dated Mar. 13, 2023. cited by applicant
U.S. Appl. No. 17/105,259, filed Nov. 25, 2020, Notice of Allowance dated Jul. 20, 2022. cited by applicant
U.S. Appl. No. 17/105,310, filed Nov. 25, 2020 Non-Final Office Action dated Feb. 22, 2023. cited by applicant
U.S. Appl. No. 17/105,310, filed Nov. 25, 2020 Notice of Allowance dated Aug. 2, 2023. cited by applicant
U.S. Appl. No. 17/357,186, filed Jun. 24, 2021 Non Final Office Action dated May 30, 2023. cited by applicant
U.S. Appl. No. 17/357,186, filed Jun. 24, 2021 Notice of Allowance dated Aug. 23, 2023. cited by applicant
U.S. Appl. No. 17/357,186, filed Jun. 24, 2021 Restriction Requirement dated Mar. 7, 2023. cited by applicant
U.S. Appl. No. 17/357,561, filed Jun. 24, 2021 Non-Final Office Action dated Aug. 11, 2022. cited by applicant
U.S. Appl. No. 17/357,561, filed Jun. 24, 2021 Notice of Allowance dated Dec. 9, 2022. cited by applicant
U.S. Appl. No. 17/371,993, filed Jul. 9, 2021 Non-Final Office Action dated Jul. 12, 2022. cited by applicant
U.S. Appl. No. 17/371,993, filed Jul. 9, 2021 Notice of Allowance dated Nov. 3, 2022. cited by applicant
U.S. Appl. No. 17/392,002, filed Aug. 2, 2021, Corrected Notice of Allowability dated Feb. 23, 2023. cited by applicant
U.S. Appl. No. 17/392,002, filed Aug. 2, 2021, Non-Final Office Action dated Sep. 12, 2022. cited by applicant
U.S. Appl. No. 17/392,002, filed Aug. 2, 2021, Notice of Allowance dated Jan. 19, 2023. cited by applicant
U.S. Appl. No. 17/484,960, filed Sep. 24, 2021 Non-Final Office Action dated Oct. 5, 2023. cited by applicant

U.S. Appl. No. 17/500,678, filed Oct. 13, 2021 Advisory Action dated Dec. 7, 2023. cited by applicant

Mayoral et al. Fiber Optic Sensors for Vital Signs Monitoring. A Review of Its Practicality in the Health Field. Biosensors (Basel). Feb. 23, 2021;11(2):58. doi: 10.3390/bios11020058. cited by applicant

U.S. Appl. No. 15/947,267, filed Apr. 6, 2018 Notice of Allowance dated Jul. 16, 2024. cited by applicant

U.S. Appl. No. 17/500,678, filed Oct. 13, 2021 Final Office Action dated Aug. 27, 2024. cited by applicant

U.S. Appl. No. 17/569,350, filed Jan. 5, 2022 Advisory Action dated Jul. 12, 2024. cited by applicant

U.S. Appl. No. 17/728,802, filed Apr. 25, 2022 Non-Final Office Action dated Aug. 28, 2024. cited by applicant

U.S. Appl. No. 17/747,903, filed May 18, 2022 Non-Final Office Action dated Aug. 15, 2024. cited by applicant

U.S. Appl. No. 17/852,138, filed Jun. 28, 2022 Non-Final Office Action dated Sep. 18, 2024. cited by applicant

U.S. Appl. No. 17/853,590, filed Jun. 29, 2022 Non-Final Office Action dated Oct. 17, 2024. cited by applicant

U.S. Appl. No. 17/955,019, filed Sep. 28, 2022 Non-Final Office Action dated Sep. 27, 2024. cited by applicant

U.S. Appl. No. 18/132,231, filed Apr. 7, 2023 Non-Final Office Action dated Jul. 12, 2024. cited by applicant

U.S. Appl. No. 18/132,623, filed Apr. 10, 2023, Final Office Action dated Sep. 6, 2024. cited by applicant

U.S. Appl. No. 18/607,144, filed Mar. 15, 2024 Non-Final Office Action dated Sep. 24, 2024. cited by applicant

U.S. Appl. No. 17/569,350, filed Jan. 5, 2022 Non-Final Office Action dated Aug. 12, 2024. cited by applicant

P.J. de Feyter, P. Kay, C. Disco, P.W. Serruys, "Reference chart derived from post-stent-implantation intravascular ultrasound predictors of 6-month expected restenosis on quantitative coronary angiography." Circulation. vol. 100, No. 17 (Year: 1999). cited by applicant

PCT/US2023/026581 filed Jun. 29, 2023 International Preliminary Report on Patentability dated Dec. 18, 2024. cited by applicant

U.S. Appl. No. 17/500,678, filed Oct. 13, 2021 Advisory Action dated Nov. 6, 2024. cited by applicant

U.S. Appl. No. 17/500,678, filed Oct. 13, 2021 Non-Final Office Action dated Dec. 4, 2024. cited by applicant

U.S. Appl. No. 17/569,350, filed Jan. 5, 2022 Notice of Allowance dated Nov. 7, 2024. cited by applicant

U.S. Appl. No. 17/585,219, filed Jan. 26, 2022 Restriction Requirement dated Dec. 2, 2024. cited by applicant

U.S. Appl. No. 17/971,873, filed Oct. 24, 2022 Non-Final Office Action dated Dec. 27, 2024. cited by applicant

U.S. Appl. No. 18/132,623, filed Apr. 10, 2023, Notice of Allowance dated Nov. 19, 2024. cited by applicant

U.S. Appl. No. 18/383,809, filed Oct. 25, 2023 Non-Final Office Action dated Dec. 16, 2024. cited by applicant

U.S. Appl. No. 17/484,960, filed Sep. 24, 2021 Notice of Allowance dated Apr. 12, 2024. cited by applicant

U.S. Appl. No. 17/569,350, filed Jan. 5, 2022 Final Office Action dated Apr. 23, 2024. cited by applicant

U.S. Appl. No. 17/747,903, filed May 18, 2022 Restriction Requirement dated May 28, 2024. cited by applicant

U.S. Appl. No. 17/955,019, filed Sep. 28, 2022 Restriction Requirement dated Jun. 6, 2024. cited by applicant

U.S. Appl. No. 18/079,653, filed Dec. 12, 2022 Notice of Allowance dated Jun. 4, 2024. cited by applicant

U.S. Appl. No. 18/132,623, filed Apr. 10, 2023, Non-Final Office Action dated May 3, 2024. cited by applicant

U.S. Appl. No. 17/971,873, filed Oct. 24, 2022 Non-Final Office Action dated Jun. 6, 2024. cited by applicant

U.S. Appl. No. 17/585,219, filed Jan. 26, 2022 Non-Final Office Action dated Feb. 27, 2025. cited by applicant

U.S. Appl. No. 17/728,802, filed Apr. 25, 2022 Advisory Action dated Apr. 4, 2025. cited by applicant

U.S. Appl. No. 17/728,802, filed Apr. 25, 2022 Final Office Action dated Jan. 17, 2025. cited by applicant

U.S. Appl. No. 17/747,903, filed May 18, 2022 Final Office Action dated Jan. 27, 2025. cited by applicant

U.S. Appl. No. 17/852,138, filed Jun. 28, 2022 Notice of Allowance dated Feb. 12, 2025. cited by applicant

U.S. Appl. No. 17/853,590, filed Jun. 29, 2022 Notice of Allowance dated Mar. 5, 2025. cited by applicant

U.S. Appl. No. 17/952,645, filed Sep. 26, 2022 Non-Final Office Action dated Feb. 28, 2025. cited by applicant

U.S. Appl. No. 17/955,019, filed Sep. 28, 2022 Final Office Action dated Feb. 4, 2025. cited by applicant

U.S. Appl. No. 18/132,231, filed Apr. 7, 2023 Final Office Action dated Jan. 31, 2025. cited by applicant

U.S. Appl. No. 18/607,144, filed Mar. 15, 2024 Final Office Action dated Feb. 19, 2025. cited by applicant

Ogawa, K.; Koyama, S.; Ishizawa, H.; Fujiwara, S.; Fujimoto, K. Simultaneous Measurement of Heart Sound, PulseWave and Respiration with Single Fiber Bragg Grating Sensor. In Proceedings of the 2018 IEEE International Symposium on Medical Measurements and Applications (MeMeA), Rome, Italy, Jun. 11-13, 2018. cited by applicant

U.S. Appl. No. 17/500,678, filed Oct. 13, 2021 Final Office Action dated Jun. 11, 2025. cited by applicant

U.S. Appl. No. 17/585,219, filed Jan. 26, 2022 Notice of Allowance dated Jun. 17, 2025. cited by applicant

U.S. Appl. No. 17/728,802, filed Apr. 25, 2022 Non-Final Office Action dated May 19, 2025. cited by applicant

U.S. Appl. No. 17/747,903, filed May 18, 2022 Advisory Action dated Apr. 22, 2025. cited by applicant

U.S. Appl. No. 17/955,019, filed Sep. 28, 2022 Advisory Action dated Apr. 16, 2025. cited by applicant

U.S. Appl. No. 18/132,231, filed Apr. 7, 2023 Notice of Allowance dated Apr. 10, 2025. cited by applicant

U.S. Appl. No. 18/132,891, filed Apr. 10, 2023 Restriction Requirement dated Jun. 2, 2025. cited by applicant

Primary Examiner: Turchen; Rochelle D

Attorney, Agent or Firm: Rutan & Tucker LLP

Background/Summary

PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 17/357,186, filed Jun. 24, 2021, now U.S. Pat. No. 11,883,609, which claims the benefit of priority to U.S. Provisional Application No. 63/045,667, filed Jun. 29, 2020, which is incorporated by reference in its entirety into this application.

BACKGROUND

(1) In the past, certain intravascular guidance of medical devices, such as guidewires and catheters for example, have used fluoroscopic methods for tracking tips of the medical devices and determining whether distal tips are appropriately localized in their target anatomical structures. However, such fluoroscopic methods expose patients and their attending clinicians to harmful X-ray radiation. Moreover, in some cases, the patients are exposed to potentially harmful contrast media needed for the fluoroscopic methods.

(2) More recently, electromagnetic tracking systems have been used involving stylets. Generally, electromagnetic tracking systems feature three components: a field generator, a sensor unit and control unit. The field generator uses several coils to generate a position-varying magnetic field, which is used to establish a coordinate space. Attached to the stylet, such as near a distal end (tip) of the stylet for example, the sensor unit includes small coils in which current is induced via the magnetic field. Based on the electrical properties of each coil, the position and orientation of the medical device may be determined within the coordinate space. The control unit controls the field generator and captures data from the sensor unit.

(3) Although electromagnetic tracking systems avoid line-of-sight reliance in tracking the tip of a stylet while obviating radiation exposure and potentially harmful contrast media associated with fluoroscopic methods, electromagnetic tracking systems are prone to interference. More specifically, since electromagnetic tracking systems depend on the measurement of magnetic fields produced by the field generator, these systems are subject to electromagnetic field interference, which may be caused by the presence of many different types of consumer electronics such as cellular telephones. Additionally, electromagnetic systems are subject to signal drop out, depend on an external sensor, and are defined to a limited depth range.

(4) Disclosed herein is a fiber optic shape sensing system and methods thereof, which is not subject to the disadvantages associated with electromagnetic tracking systems as described above and is capable in its current form of providing confirmation of tip placement or information passed/interpreted as an electrical signal. Further, inventive systems configured to and methods of determining an orientation of a stylet advancing within a patient's vasculature based at least partially on the information passed/interpreted as an electrical signal are disclosed. Additionally, such inventive systems and method that may perform or include operations of displaying information received from the fiber optic shape sensing system based on the determined orientation are disclosed.

SUMMARY

(5) Briefly summarized, embodiments disclosed herein are directed to systems, apparatus and methods for obtaining three-dimensional (3D) information (reflected light) corresponding to a

trajectory and/or shape of a medical instrument, such as a catheter, a guidewire, or a stylet, during advancement through a vasculature of a patient, determining an orientation of the medical instrument relative to the patient, and generating and rendering a two-dimensional (2D) display of the medical instrument in real-time in accordance with the determined orientation.

(6) More particularly, in some embodiments, the medical instrument includes a multi-core optical fiber, with each core fiber of the multi-core optical fiber is configured with an array of sensors (reflective gratings), which are spatially distributed over a prescribed length of the core fiber to generally sense external strain those regions of the core fiber occupied by the sensor. The multi-core optical fiber is configured to receive broadband light from a console during advancement through the vasculature of a patient, where the broadband light propagates along at least a partial distance of the multi-core optical fiber toward the distal end. Given that each sensor positioned along the same core fiber is configured to reflect light of a different, specific spectral width, the array of sensors enables distributed measurements throughout the prescribed length of the multi-core optical fiber. These distributed measurements may include wavelength shifts having a correlation with strain experienced by the sensor.

(7) The reflected light from the sensors (reflective gratings) within each core fiber of the multi-core optical fiber is returned from the medical instrument for processing by the console. The physical state of the medical instrument may be ascertained based on analytics of the wavelength shifts of the reflected light. For example, the strain caused through bending of the medical instrument, and hence angular modification of each core fiber, causes different degrees of deformation. The different degrees of deformation alters the shape of the sensors (reflective grating) positioned on the core fiber, which may cause variations (shifts) in the wavelength of the reflected light from the sensors positioned on each core fiber within the multi-core optical fiber, as shown in FIGS. 2, 7 and 10.

(8) From this wavelength shifting, logic within the console may determine the physical state of the medical instrument (e.g., shape). Additionally, based on the wavelength shifting and anatomical constraints of the patient's body, the logic within the console may determine an orientation of the medical instrument relative to a known reference frame of the patient's body. Subsequently, the logic within the console is configured to generate a 2D display of the physical state of the medical instrument using the determined orientation of the medical instrument, where the reference frame of the 2D display mirrors the reference frame of the patient's body. Thus, the 2D display depicts a current physical state of the catheter displayed in such a manner as to be representative of the anatomical positioning and orientation relative to the patient's body, which enables a clinician to perceive the advancement of the medical instrument in an anatomically proper manner.

(9) Specific embodiments of the disclosure include utilization of a stylet featuring a multi-core optical fiber and a conductive medium that collectively operate for tracking placement of a catheter or other medical device within a body of a patient. The stylet is configured to return information for use in identifying its physical state (e.g., shape length, shape, and/or form) of (i) a portion of the stylet (e.g., tip, segment of stylet, etc.) or a portion of a catheter inclusive of at least a portion of the stylet (e.g., tip, segment of catheter, etc.) or (ii) the entirety or a substantial portion of the stylet or catheter within the body of a patient (hereinafter, described as the “physical state of the stylet” or the “physical state of the catheter”). According to one embodiment of the disclosure, the returned information may be obtained from reflected light signals of different spectral widths, where each reflected light signal corresponds to a portion of broadband incident light propagating along a core of the multi-core optical fiber (hereinafter, “core fiber”) that is reflected back over the core fiber by a particular sensor located on the core fiber. One illustrative example of the returned information may pertain to a change in signal characteristics of the reflected light signal returned from the sensor, where wavelength shift is correlated to (mechanical) strain on the core fiber.

(10) In some embodiments, the stylet includes a multi-core optical fiber, where each core fiber utilizes a plurality of sensors and each sensor is configured to reflect a different spectral range of

the incident light (e.g., different light frequency range). Based on the type and degree of strain asserted on each core fiber, the sensors associated with that core fiber may alter (shift) the wavelength of the reflected light to convey the type and degree of stain on that core fiber at those locations of the stylet occupied by the sensors. The sensors are spatially distributed at various locations of the core fiber between a proximal end and a distal end of the stylet so that shape sensing of the stylet may be conducted based on analytics of the wavelength shifts. Herein, the shape sensing functionality is paired with the ability to simultaneously pass an electrical signal through the same member (stylet) through conductive medium included as part of the stylet.

(11) More specifically, in some embodiments each core fiber of the multi-core optical fiber is configured with an array of sensors, which are spatially distributed over a prescribed length of the core fiber to generally sense external strain those regions of the core fiber occupied by the sensor. Given that each sensor positioned along the same core fiber is configured to reflect light of a different, specific spectral width, the array of sensors enables distributed measurements throughout the prescribed length of the multi-core optical fiber. These distributed measurements may include wavelength shifts having a correlation with strain experienced by the sensor.

(12) According to one embodiment of the disclosure, each sensor may operate as a reflective grating such as a fiber Bragg grating (FBG), namely an intrinsic sensor corresponding to a permanent, periodic refractive index change inscribed into the core fiber. Stated differently, the sensor operates as a light reflective mirror for a specific spectral width (e.g., a specific wavelength or specific range of wavelengths). As a result, as broadband incident light is supplied by an optical light source and propagates through a particular core fiber, upon reaching a first sensor of the distributed array of sensors for that core fiber, light of a prescribed spectral width associated with the first sensor is reflected back to an optical receiver within a console, including a display and the optical light source. The remaining spectrum of the incident light continues propagation through the core fiber toward a distal end of the stylet. The remaining spectrum of the incident light may encounter other sensors from the distributed array of sensors, where each of these sensors is fabricated to reflect light with different specific spectral widths to provide distributed measurements, as described above.

(13) During operation, multiple light reflections (also referred to as “reflected light signals”) are returned to the console from each of the plurality of core fibers of the multi-core optical fiber. Each reflected light signal may be uniquely associated with a different spectral width. Information associated with the reflected light signals may be used to determine a three-dimensional representation of the physical state of the stylet within the body of a patient. Herein, the core fibers are spatially separated with the cladding of the multi-mode optical fiber and each core fiber is configured to separately return light of different spectral widths (e.g., specific light wavelength or a range of light wavelengths) reflected from the distributed array of sensors fabricated in each of the core fibers. A comparison of detected shifts in wavelength of the reflected light returned by a center core fiber (operating as a reference) and the surrounding, periphery core fibers may be used to determine the physical state of the stylet.

(14) During vasculature insertion and advancement of the catheter, the clinician may rely on the console to visualize a current physical state (e.g., shape) of a catheter guided by the stylet to avoid potential path deviations. As the periphery core fibers reside at spatially different locations within the cladding of the multi-mode optical fiber, changes in angular orientation (such as bending with respect to the center core fiber, etc.) of the stylet imposes different types (e.g., compression or tension) and degrees of strain on each of the periphery core fibers as well as the center core fiber. The different types and/or degree of strain may cause the sensors of the core fibers to apply different wavelength shifts, which can be measured to extrapolate the physical state of the stylet (catheter).

(15) In order for the clinician to rely on the console to visualize the current physical state of the catheter, the console displays a visual representation, typically a 2D image, of the physical state of

the catheter based on the reflected light. In generating the 2D image of the physical state of the catheter, the image must be displayed in such a manner as to be representative of the anatomical positioning relative to the patient's body. Thus, following analytics of the reflected light, logic of the console determines an orientation of the stylet relative to a known frame of reference of the patient's body. The logic then generates and renders a 2D display of the physical state of the catheter and a representation of the patient's body, with the display properly orienting the catheter relative to the patient's body based on the determined orientation of the stylet.

(16) Although the reflected light provides a wealth of information capable of enabling extrapolation of the physical state of the stylet (catheter), such information lacks context as to the orientation of the stylet advancing within the patient's body. For instance, the wavelength shifts of the reflected light may indicate different degrees of deformation of each core fiber, which collectively may indicate a curvature of the stylet; however, the indication of such a curvature lacks context as to an orientation or direction with respect to the patient's body. Thus, the wavelength shifts of the reflected light, alone, do not enable the logic to generate a 2D display of the physical state of the catheter that is properly oriented within a visual representation of the patient's body.

(17) Embodiments of the disclosure describe how the logic of the console utilizes the known anatomical constraints of the human body, a known insertion site of the stylet within the patient's body, and/or the wavelength shifts of the reflected light to determine an orientation of the stylet with respect to the patient's body. Additional embodiments further describe the generation of displays that illustrate the catheter within the patient's body based on the orientation of the stylet disposed within the catheter. Typically, such displays are 2D; however, as the wavelength shifts of the reflected light provide 3D information, generation of 3D displays has also been considered.

(18) Some embodiments of the invention disclose a method for placing a medical device into a body of a patient comprising providing a broadband incident light signal to each of a plurality of reflective gratings distributed along a length of each of a plurality of core fibers of a multi-core optical fiber, the plurality of core fibers being spatially distributed to experience different degrees of strain, receiving reflected light signals of different spectral widths of the broadband incident light by each of the plurality of reflective gratings, processing the reflected light signals received from each of the plurality of reflective gratings associated with the plurality of core fibers to determine (i) a physical state of the multi-core optical fiber relating to the medical device including the multi-core optical fiber, and (ii) an orientation of the multi-core optical fiber relative to a reference frame of the body.

(19) Some of the embodiments further include generating a display illustrating the physical state of the multi-core optical fiber based at least on the orientation determined during processing of the reflected light. Additionally, the display may be a two-dimensional representation of the physical state of the multi-core optical fiber in accordance with the orientation determined during processing of the reflected light.

(20) In some embodiments, the physical state of the multi-core optical fiber relating to the medical device includes one or more of a length, a shape or a form as currently possessed by the multi-core optical fiber. In further embodiments, the different types of strain include compression and tension.

(21) In some instances, determining the orientation of the multi-core optical fiber relative to the reference frame of the body includes establishing the reference frame of the body utilizing a coordinate system, establishing an initial direction of advancement along a first axis of the coordinate system for the multi-core optical fiber based on the multi-core optical fiber entering the body at a known insertion site, correlating initial reflected light signals to the initial direction of advancement along the first axis of the coordinate system, detecting a curve in the advancement of the multi-core optical fiber based on processing of the reflected light signals, and correlating reflected light signals corresponding to the curve in the advancement with a second direction of advancement along a second axis of the coordinate system, wherein the orientation is defined by (i) the initial reflected light signals correlated with the initial direction of advancement along the first

axis of the coordinate system, and (ii) the reflected light signals corresponding to the curve in the advancement correlated with the second direction of advancement along the second axis of the coordinate system. The medical device may include an elongated shape and is inserted into a vasculature of the body of the patient. In some embodiments, the medical device is a stylet removably inserted into a lumen of a catheter assembly for placement of a distal tip of the catheter assembly in a superior vena cava of the vasculature. Further, at least two of the plurality of core fibers may be configured to experience different types of strain in response to changes in an orientation of the multi-core optical fiber. In some embodiments, each reflective grating of the plurality of reflective gratings alters its reflected light signal by applying a wavelength shift dependent on a strain experienced by the reflective grating.

(22) Some embodiments of the invention disclose non-transitory computer-readable medium having stored thereon logic that, when executed by the one or more processors, causes operations including providing a broadband incident light signal to each of a plurality of reflective gratings distributed along a length of each of a plurality of core fibers of a multi-core optical fiber, the plurality of core fibers being spatially distributed to experience different degrees of strain, receiving reflected light signals of different spectral widths of the broadband incident light by each of the plurality of reflective gratings, processing the reflected light signals received from each of the plurality of reflective gratings associated with the plurality of core fibers to determine (i) a physical state of the multi-core optical fiber relating to the medical device including the multi-core optical fiber, and (ii) an orientation of the multi-core optical fiber relative to a reference frame of the body.

(23) Some of the embodiments further include generating a display illustrating the physical state of the multi-core optical fiber based at least on the orientation determined during processing of the reflected light. Additionally, the display may be a two-dimensional representation of the physical state of the multi-core optical fiber in accordance with the orientation determined during processing of the reflected light.

(24) In some embodiments, the physical state of the multi-core optical fiber relating to the medical device includes one or more of a length, a shape or a form as currently possessed by the multi-core optical fiber. In further embodiments, the different types of strain include compression and tension.

(25) In some instances, determining the orientation of the multi-core optical fiber relative to the reference frame of the body includes establishing the reference frame of the body utilizing a coordinate system, establishing an initial direction of advancement along a first axis of the coordinate system for the multi-core optical fiber based on the multi-core optical fiber entering the body at a known insertion site, correlating initial reflected light signals to the initial direction of advancement along the first axis of the coordinate system, detecting a curve in the advancement of the multi-core optical fiber based on processing of the reflected light signals, and correlating reflected light signals corresponding to the curve in the advancement with a second direction of advancement along a second axis of the coordinate system, wherein the orientation is defined by (i) the initial reflected light signals correlated with the initial direction of advancement along the first axis of the coordinate system, and (ii) the reflected light signals corresponding to the curve in the advancement correlated with the second direction of advancement along the second axis of the coordinate system. The medical device may include an elongated shape and is inserted into a vasculature of the body of the patient. In some embodiments, the medical device is a stylet removably inserted into a lumen of a catheter assembly for placement of a distal tip of the catheter assembly in a superior vena cava of the vasculature. Further, at least two of the plurality of core fibers may be configured to experience different types of strain in response to changes in an orientation of the multi-core optical fiber. In some embodiments, each reflective grating of the plurality of reflective gratings alters its reflected light signal by applying a wavelength shift dependent on a strain experienced by the reflective grating.

(26) Some embodiments of the invention disclose a medical device comprising a multi-core optical

fiber having a plurality of core fibers, each of the plurality of core fibers including a plurality of sensors distributed along a longitudinal length of a corresponding core fiber and each sensor of the plurality of sensors being configured to (i) reflect a light signal of a different spectral width based on received incident light, and (ii) change a characteristic of the reflected light signal for use in determining a physical state of the multi-core optical fiber, and a console including one or more processors and a non-transitory computer-readable medium having stored thereon logic that, when executed by the one or more processors, causing performance of certain operations. These certain operations may include providing a broadband incident light signal to each of a plurality of reflective gratings distributed along a length of each of a plurality of core fibers of a multi-core optical fiber, the plurality of core fibers being spatially distributed to experience different degrees of strain, receiving reflected light signals of different spectral widths of the broadband incident light by each of the plurality of reflective gratings, processing the reflected light signals received from each of the plurality of reflective gratings associated with the plurality of core fibers to determine (i) a physical state of the multi-core optical fiber relating to the medical device including the multi-core optical fiber, and (ii) an orientation of the multi-core optical fiber relative to a reference frame of the body.

(27) Some of the embodiments further include generating a display illustrating the physical state of the multi-core optical fiber based at least on the orientation determined during processing of the reflected light. Additionally, the display may be a two-dimensional representation of the physical state of the multi-core optical fiber in accordance with the orientation determined during processing of the reflected light.

(28) In some embodiments, the physical state of the multi-core optical fiber relating to the medical device includes one or more of a length, a shape or a form as currently possessed by the multi-core optical fiber. In further embodiments, the different types of strain include compression and tension.

(29) In some instances, determining the orientation of the multi-core optical fiber relative to the reference frame of the body includes establishing the reference frame of the body utilizing a coordinate system, establishing an initial direction of advancement along a first axis of the coordinate system for the multi-core optical fiber based on the multi-core optical fiber entering the body at a known insertion site, correlating initial reflected light signals to the initial direction of advancement along the first axis of the coordinate system, detecting a curve in the advancement of the multi-core optical fiber based on processing of the reflected light signals, and correlating reflected light signals corresponding to the curve in the advancement with a second direction of advancement along a second axis of the coordinate system, wherein the orientation is defined by (i) the initial reflected light signals correlated with the initial direction of advancement along the first axis of the coordinate system, and (ii) the reflected light signals corresponding to the curve in the advancement correlated with the second direction of advancement along the second axis of the coordinate system. The medical device may include an elongated shape and is inserted into a vasculature of the body of the patient. In some embodiments, the medical device is a stylet removably inserted into a lumen of a catheter assembly for placement of a distal tip of the catheter assembly in a superior vena cava of the vasculature. Further, at least two of the plurality of core fibers may be configured to experience different types of strain in response to changes in an orientation of the multi-core optical fiber. In some embodiments, each reflective grating of the plurality of reflective gratings alters its reflected light signal by applying a wavelength shift dependent on a strain experienced by the reflective grating.

(30) These and other features of the concepts provided herein will become more apparent to those of skill in the art in view of the accompanying drawings and following description, which disclose particular embodiments of such concepts in greater detail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the disclosure are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings, in which like references indicate similar elements and in which:
- (2) FIG. 1A is an illustrative embodiment of a medical instrument monitoring system including a medical instrument with optic shape sensing and fiber optic-based oximetry capabilities in accordance with some embodiments;
- (3) FIG. 1B is an alternative illustrative embodiment of the medical instrument monitoring system **100** in accordance with some embodiments;
- (4) FIG. 2 is an exemplary embodiment of a structure of a section of the multi-core optical fiber included within the stylet **120** of FIG. 1A in accordance with some embodiments;
- (5) FIG. 3A is a first exemplary embodiment of the stylet of FIG. 1A supporting both an optical and electrical signaling in accordance with some embodiments;
- (6) FIG. 3B is a cross sectional view of the stylet of FIG. 3A in accordance with some embodiments;
- (7) FIG. 4A is a second exemplary embodiment of the stylet of FIG. 1B in accordance with some embodiments;
- (8) FIG. 4B is a cross sectional view of the stylet of FIG. 4A in accordance with some embodiments;
- (9) FIG. 5A is an elevation view of a first illustrative embodiment of a catheter including integrated tubing, a diametrically disposed septum, and micro-lumens formed within the tubing and septum in accordance with some embodiments;
- (10) FIG. 5B is a perspective view of the first illustrative embodiment of the catheter of FIG. 5A including core fibers installed within the micro-lumens in accordance with some embodiments;
- (11) FIGS. 6A-6B are flowcharts of the methods of operations conducted by the medical instrument monitoring system of FIGS. 1A-1B to achieve optic 3D shape sensing in accordance with some embodiments;
- (12) FIG. 7 is an exemplary embodiment of the medical instrument monitoring system of FIG. 1B during operation and insertion of the catheter into a patient in accordance with some embodiments;
- (13) FIGS. 8A-8B are perspective illustrations of 3D catheter images within planes of 3D Cartesian coordinate systems in accordance with some embodiments in accordance with some embodiments;
- (14) FIGS. 9A-9B illustrate a flowchart of a first embodiment of operations performed by logic of the medical instrument monitoring system of FIG. 1A in accordance with some embodiments;
- (15) FIG. 10 is a second exemplary embodiment of the medical instrument monitoring system of FIG. 1A during operation and insertion of the catheter into a patient in accordance with some embodiments; and
- (16) FIGS. 11A-11B illustrate a flowchart of a second embodiment of operations performed by logic of the medical instrument monitoring system of FIG. 1A in accordance with some embodiments.

DETAILED DESCRIPTION

- (17) Before some particular embodiments are disclosed in greater detail, it should be understood that the particular embodiments disclosed herein do not limit the scope of the concepts provided herein. It should also be understood that a particular embodiment disclosed herein can have features that can be readily separated from the particular embodiment and optionally combined with or substituted for features of any of a number of other embodiments disclosed herein.
- (18) Regarding terms used herein, it should also be understood the terms are for the purpose of describing some particular embodiments, and the terms do not limit the scope of the concepts provided herein. Ordinal numbers (e.g., first, second, third, etc.) are generally used to distinguish or identify different features or steps in a group of features or steps, and do not supply a serial or

numerical limitation. For example, “first,” “second,” and “third” features or steps need not necessarily appear in that order, and the particular embodiments including such features or steps need not necessarily be limited to the three features or steps. Labels such as “left,” “right,” “top,” “bottom,” “front,” “back,” and the like are used for convenience and are not intended to imply, for example, any particular fixed location, orientation, or direction. Instead, such labels are used to reflect, for example, relative location, orientation, or directions. Singular forms of “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

(19) With respect to “proximal,” a “proximal portion” or a “proximal end portion” of, for example, a probe disclosed herein includes a portion of the probe intended to be near a clinician when the probe is used on a patient. Likewise, a “proximal length” of, for example, the probe includes a length of the probe intended to be near the clinician when the probe is used on the patient. A “proximal end” of, for example, the probe includes an end of the probe intended to be near the clinician when the probe is used on the patient. The proximal portion, the proximal end portion, or the proximal length of the probe can include the proximal end of the probe; however, the proximal portion, the proximal end portion, or the proximal length of the probe need not include the proximal end of the probe. That is, unless context suggests otherwise, the proximal portion, the proximal end portion, or the proximal length of the probe is not a terminal portion or terminal length of the probe.

(20) With respect to “distal,” a “distal portion” or a “distal end portion” of, for example, a probe disclosed herein includes a portion of the probe intended to be near or in a patient when the probe is used on the patient. Likewise, a “distal length” of, for example, the probe includes a length of the probe intended to be near or in the patient when the probe is used on the patient. A “distal end” of, for example, the probe includes an end of the probe intended to be near or in the patient when the probe is used on the patient. The distal portion, the distal end portion, or the distal length of the probe can include the distal end of the probe; however, the distal portion, the distal end portion, or the distal length of the probe need not include the distal end of the probe. That is, unless context suggests otherwise, the distal portion, the distal end portion, or the distal length of the probe is not a terminal portion or terminal length of the probe.

(21) The term “logic” may be representative of hardware, firmware or software that is configured to perform one or more functions. As hardware, the term logic may refer to or include circuitry having data processing and/or storage functionality. Examples of such circuitry may include, but are not limited or restricted to a hardware processor (e.g., microprocessor, one or more processor cores, a digital signal processor, a programmable gate array, a microcontroller, an application specific integrated circuit “ASIC”, etc.), a semiconductor memory, or combinatorial elements.

(22) Additionally, or in the alternative, the term logic may refer to or include software such as one or more processes, one or more instances, Application Programming Interface(s) (API), subroutine(s), function(s), applet(s), servlet(s), routine(s), source code, object code, shared library/dynamic link library (dll), or even one or more instructions. This software may be stored in any type of a suitable non-transitory storage medium, or transitory storage medium (e.g., electrical, optical, acoustical or other form of propagated signals such as carrier waves, infrared signals, or digital signals). Examples of a non-transitory storage medium may include, but are not limited or restricted to a programmable circuit; non-persistent storage such as volatile memory (e.g., any type of random access memory “RAM”); or persistent storage such as non-volatile memory (e.g., read-only memory “ROM”, power-backed RAM, flash memory, phase-change memory, etc.), a solid-state drive, hard disk drive, an optical disc drive, or a portable memory device. As firmware, the logic may be stored in persistent storage.

(23) Referring to FIG. 1A, an illustrative embodiment of a medical instrument monitoring system including a medical instrument with optic shape sensing and fiber optic-based oximetry capabilities is shown in accordance with some embodiments. As shown, the system **100** generally includes a console **110** and a stylet assembly **119** communicatively coupled to the console **110**. For this

embodiment, the stylet assembly **119** includes an elongate probe (e.g., stylet) **120** on its distal end **122** and a console connector **133** on its proximal end **124**, where the stylet **120** is configured to advance within a patient vasculature either through, or in conjunction with, a catheter **195**. The console connector **133** enables the stylet assembly **119** to be operably connected to the console **110** via an interconnect **145** including one or more optical fibers **147** (hereinafter, “optical fiber(s)”) and a conductive medium terminated by a single optical/electric connector **146** (or terminated by dual connectors. Herein, the connector **146** is configured to engage (mate) with the console connector **133** to allow for the propagation of light between the console **110** and the stylet assembly **119** as well as the propagation of electrical signals from the stylet **120** to the console **110**.

(24) An exemplary implementation of the console **110** includes a processor **160**, a memory **165**, a display **170** and optical logic **180**, although it is appreciated that the console **110** can take one of a variety of forms and may include additional components (e.g., power supplies, ports, interfaces, etc.) that are not directed to aspects of the disclosure. An illustrative example of the console **110** is illustrated in U.S. Publication No. 2019/0237902, the entire contents of which are incorporated by reference herein. The processor **160**, with access to the memory **165** (e.g., non-volatile memory or non-transitory, computer-readable medium), is included to control functionality of the console **110** during operation. As shown, the display **170** may be a liquid crystal diode (LCD) display integrated into the console **110** and employed as a user interface to display information to the clinician, especially during a catheter placement procedure (e.g., cardiac catheterization). In another embodiment, the display **170** may be separate from the console **110**. Although not shown, a user interface is configured to provide user control of the console **110**.

(25) For both of these embodiments, the content depicted by the display **170** may change according to which mode the stylet **120** is configured to operate: optical, TLS, ECG, or another modality. In TLS mode, the content rendered by the display **170** may constitute a two-dimensional (2D) or three-dimensional (3D) representation of the physical state (e.g., length, shape, form, and/or orientation) of the stylet **120** computed from characteristics of reflected light signals **150** returned to the console **110**. The reflected light signals **150** constitute light of a specific spectral width of broadband incident light **155** reflected back to the console **110**. According to one embodiment of the disclosure, the reflected light signals **150** may pertain to various discrete portions (e.g., specific spectral widths) of broadband incident light **155** transmitted from and sourced by the optical logic **180**, as described below

(26) According to one embodiment of the disclosure, an activation control **126**, included on the stylet assembly **119**, may be used to set the stylet **120** into a desired operating mode and selectively alter operability of the display **170** by the clinician to assist in medical device placement. For example, based on the modality of the stylet **120**, the display **170** of the console **110** can be employed for optical modality-based guidance during catheter advancement through the vasculature or TLS modality to determine the physical state (e.g., length, form, shape, orientation, etc.) of the stylet **120**. In one embodiment, information from multiple modes, such as optical, TLS or ECG for example, may be displayed concurrently (e.g., at least partially overlapping in time).

(27) Referring still to FIG. 1A, the optical logic **180** is configured to support operability of the stylet assembly **119** and enable the return of information to the console **110**, which may be used to determine the physical state associated with the stylet **120** along with monitored electrical signals such as ECG signaling via an electrical signaling logic **181** that supports receipt and processing of the received electrical signals from the stylet **120** (e.g., ports, analog-to-digital conversion logic, etc.). The physical state of the stylet **120** may be based on changes in characteristics of the reflected light signals **150** received at the console **110** from the stylet **120**. The characteristics may include shifts in wavelength caused by strain on certain regions of the core fibers integrated within an optical fiber core **135** positioned within or operating as the stylet **120**, as shown below. As discussed herein, the optical fiber core **135** may be comprised of core fibers **137.sub.1-137.sub.M** ($M=1$ for a single core, and $M \geq 2$ for a multi-core), where the core fibers **137.sub.1-137.sub.M** may

collectively be referred to as core fiber(s) **137**. Unless otherwise specified or the instant embodiment requires an alternative interpretation, embodiments discussed herein will refer to a multi-core optical fiber **135**. From information associated with the reflected light signals **150**, the console **110** may determine (through computation or extrapolation of the wavelength shifts) the physical state of the stylet **120**, and also that of the catheter **195** configured to receive the stylet **120**.

(28) According to one embodiment of the disclosure, as shown in FIG. **1A**, the optical logic **180** may include a light source **182** and an optical receiver **184**. The light source **182** is configured to transmit the incident light **155** (e.g., broadband) for propagation over the optical fiber(s) **147** included in the interconnect **145**, which are optically connected to the multi-core optical fiber core **135** within the stylet **120**. In one embodiment, the light source **182** is a tunable swept laser, although other suitable light sources can also be employed in addition to a laser, including semi-coherent light sources, LED light sources, etc.

(29) The optical receiver **184** is configured to: (i) receive returned optical signals, namely reflected light signals **150** received from optical fiber-based reflective gratings (sensors) fabricated within each core fiber of the multi-core optical fiber **135** deployed within the stylet **120**, and (ii) translate the reflected light signals **150** into reflection data (from repository **192**), namely data in the form of electrical signals representative of the reflected light signals including wavelength shifts caused by strain. The reflected light signals **150** associated with different spectral widths may include reflected light signals **151** provided from sensors positioned in the center core fiber (reference) of the multi-core optical fiber **135** and reflected light signals **152** provided from sensors positioned in the periphery core fibers of the multi-core optical fiber **135**, as described below. Herein, the optical receiver **184** may be implemented as a photodetector, such as a positive-intrinsic-negative “PIN” photodiode, avalanche photodiode, or the like.

(30) As shown, both the light source **182** and the optical receiver **184** are operably connected to the processor **160**, which governs their operation. Also, the optical receiver **184** is operably coupled to provide the reflection data (from repository **192**) to the memory **165** for storage and processing by reflection data classification logic **190**. The reflection data classification logic **190** may be configured to: (i) identify which core fibers pertain to which of the received reflection data (from repository **192**) and (ii) segregate the reflection data stored with a repository **192** provided from reflected light signals **150** pertaining to similar regions of the stylet **120** or spectral widths into analysis groups. The reflection data for each analysis group is made available to shape sensing logic **194** for analytics.

(31) According to one embodiment of the disclosure, the shape sensing logic **194** is configured to compare wavelength shifts measured by sensors deployed in each periphery core fiber at the same measurement region of the stylet **120** (or same spectral width) to the wavelength shift at a center core fiber of the multi-core optical fiber **135** positioned along central axis and operating as a neutral axis of bending. From these analytics, the shape sensing logic **194** may determine the shape the core fibers have taken in 3D space and may further determine the current physical state of the catheter **195** in 3D space for rendering on the display **170**.

(32) According to one embodiment of the disclosure, the shape sensing logic **194** may generate a rendering of the current physical state of the stylet **120** (and potentially the catheter **195**), based on heuristics or run-time analytics. For example, the shape sensing logic **194** may be configured in accordance with machine-learning techniques to access a data store (library) with pre-stored data (e.g., images, etc.) pertaining to different regions of the stylet **120** (or catheter **195**) in which reflected light from core fibers have previously experienced similar or identical wavelength shifts. From the pre-stored data, the current physical state of the stylet **120** (or catheter **195**) may be rendered. Alternatively, as another example, the shape sensing logic **194** may be configured to determine, during run-time, changes in the physical state of each region of the multi-core optical fiber **135** based on at least: (i) resultant wavelength shifts experienced by different core fibers

within the optical fiber **135**, and (ii) the relationship of these wavelength shifts generated by sensors positioned along different periphery core fibers at the same cross-sectional region of the multi-core optical fiber **135** to the wavelength shift generated by a sensor of the center core fiber at the same cross-sectional region. It is contemplated that other processes and procedures may be performed to utilize the wavelength shifts as measured by sensors along each of the core fibers within the multi-core optical fiber **135** to render appropriate changes in the physical state of the stylet **120** (and/or catheter **195**), especially to enable guidance of the stylet **120**, when positioned at a distal tip of the catheter **195**, within the vasculature of the patient and at a desired destination within the body.

(33) The console **110** may further include electrical signaling logic **181**, which is positioned to receive one or more electrical signals from the stylet **120**. The stylet **120** is configured to support both optical connectivity as well as electrical connectivity. The electrical signaling logic **181** receives the electrical signals (e.g., ECG signals) from the stylet **120** via the conductive medium. The electrical signals may be processed by electrical signal logic **196**, executed by the processor **160**, to determine ECG waveforms for display.

(34) Additionally, the console **110** includes a fluctuation logic **198** that is configured to analyze at least a subset of the wavelength shifts measured by sensors deployed in each of the core fibers **137**. In particular, the fluctuation logic **198** is configured to analyze wavelength shifts measured by sensors of core fibers **137**, where such corresponds to an analysis of the fluctuation of the distal tip of the stylet **120** (or “tip fluctuation analysis”). In some embodiments, the fluctuation logic **198** measures analyzes the wavelength shifts measured by sensors at a distal end of the core fibers **137**. The tip fluctuation analysis includes at least a correlation of detected movements of the distal tip of the stylet **120** (or other medical device or instrument) with experiential knowledge comprising previously detected movements (fluctuations), and optionally, other current measurements such as ECG signals. The experiential knowledge may include previously detected movements in various locations within the vasculature (e.g., SVC, Inferior Vena Cava (IVC), right atrium, azygos vein, other blood vessels such as arteries and veins) under normal, healthy conditions and in the presence of defects (e.g., vessel constriction, vasospasm, vessel occlusion, etc.). Thus, the tip fluctuation analysis may result in a confirmation of tip location and/or detection of a defect affecting a blood vessel.

(35) It should be noted that the fluctuation logic **198** need not perform the same analyses as the shape sensing logic **194**. For instance, the shape sensing logic **194** determines a 3D shape of the stylet **120** by comparing wavelength shifts in outer core fibers of a multi-core optical fiber to a center, reference core fiber. The fluctuation logic **198** may instead correlate the wavelength shifts to previously measured wavelength shifts and optionally other current measurements without distinguishing between wavelength shifts of outer core fibers and a center, reference core fiber as the tip fluctuation analysis need not consider direction or shape within a 3D space.

(36) In some embodiments, e.g., those directed at tip location confirmation, the analysis of the fluctuation logic **198** may utilize electrical signals (e.g., ECG signals) measured by the electrical signaling logic **181**. For example, the fluctuation logic **198** may compare the movements of a subsection of the stylet **120** (e.g., the distal tip) with electrical signals indicating impulses of the heart (e.g., the heartbeat). Such a comparison may reveal whether the distal tip is within the SVC or the right atrium based on how closely the movements correspond to a rhythmic heartbeat.

(37) In various embodiments, a display and/or alert may be generated based on the fluctuation analysis. For instance, the fluctuation logic **198** may generate a graphic illustrating the detected fluctuation compared to previously detected tip fluctuations and/or the anatomical movements of the patient body such as rhythmic pulses of the heart and/or expanding and contracting of the lungs. In one embodiment, such a graphic may include a dynamic visualization of the present medical device moving in accordance with the detected fluctuations adjacent to a secondary medical device moving in accordance with previously detected tip fluctuations. In some embodiments, the location

of a subsection of the medical device may be obtained from the shape sensing logic **194** and the dynamic visualization may be location-specific (e.g., such that the previously detected fluctuations illustrate expected fluctuations for the current location of the subsection). In alternative embodiments, the dynamic visualization may illustrate a comparison of the dynamic movements of the subsection to one or more subsections moving in accordance with previously detected fluctuations of one or more defects affecting the blood vessel.

(38) According to one embodiment of the disclosure, the fluctuation logic **198** may determine whether movements of one or more subsections of the stylet **120** indicate a location of a particular subsection of the stylet **120** or a defect affecting a blood vessel and, as a result, of the catheter **195**, based on heuristics or run-time analytics. For example, the fluctuation logic **198** may be configured in accordance with machine-learning techniques to access a data store (library) with pre-stored data (e.g., experiential knowledge of previously detected tip fluctuation data, etc.) pertaining to different regions (subsections) of the stylet **120**. Specifically, such an embodiment may include processing of a machine-learning model trained using the experiential knowledge, where the detected fluctuations serve as input to the trained model and processing of the trained model results in a determination as to how closely the detected fluctuations correlate to one or more locations within the vasculature of the patient and/or one or more defects affecting a blood vessel.

(39) In some embodiments, the fluctuation logic **198** may be configured to determine, during run-time, whether movements of one or more subsections of the stylet **120** (and the catheter **195**) indicate a location of a particular subsection of the stylet **120** or a defect affecting a blood vessel, based on at least (i) resultant wavelength shifts experienced by the core fibers **137** within the one or more subsections, and (ii) the correlation of these wavelength shifts generated by sensors positioned along different core fibers at the same cross-sectional region of the stylet **120** (or the catheter **195**) to previously detected wavelength shifts generated by corresponding sensors in a core fiber at the same cross-sectional region. It is contemplated that other processes and procedures may be performed to utilize the wavelength shifts as measured by sensors along each of the core fibers **137** to render appropriate movements in the distal tip of the stylet **120** and/or the catheter **195**.

(40) Referring to FIG. **1B**, an alternative exemplary embodiment of a medical instrument monitoring system **100** is shown. Herein, the medical instrument monitoring system **100** features a console **110** and a medical instrument **130** communicatively coupled to the console **110**. For this embodiment, the medical instrument **130** corresponds to a catheter, which features an integrated tubing with two or more lumen extending between a proximal end **131** and a distal end **132** of the integrated tubing. The integrated tubing (sometimes referred to as “catheter tubing”) is in communication with one or more extension legs **140** via a bifurcation hub **142**. An optical-based catheter connector **144** may be included on a proximal end of at least one of the extension legs **140** to enable the catheter **130** to operably connect to the console **110** via an interconnect **145** or another suitable component. Herein, the interconnect **145** may include a connector **146** that, when coupled to the optical-based catheter connector **144**, establishes optical connectivity between one or more optical fibers **147** (hereinafter, “optical fiber(s)”) included as part of the interconnect **145** and core fibers **137** deployed within the catheter **130** and integrated into the tubing. Alternatively, a different combination of connectors, including one or more adapters, may be used to optically connect the optical fiber(s) **147** to the core fibers **137** within the catheter **130**. The core fibers **137** deployed within the catheter **130** as illustrated in FIG. **1B** include the same characteristics and perform the same functionalities as the core fibers **137** deployed within the stylet **120** of FIG. **1A**.

(41) The optical logic **180** is configured to support graphical rendering of the catheter **130**, most notably the integrated tubing of the catheter **130**, based on characteristics of the reflected light signals **150** received from the catheter **130**. The characteristics may include shifts in wavelength caused by strain on certain regions of the core fibers **137** integrated within (or along) a wall of the integrated tubing, which may be used to determine (through computation or extrapolation of the wavelength shifts) the physical state of the catheter **130**, notably its integrated tubing or a portion

of the integrated tubing such as a tip or distal end of the tubing to read fluctuations (real-time movement) of the tip (or distal end).

(42) More specifically, the optical logic **180** includes a light source **182**. The light source **182** is configured to transmit the broadband incident light **155** for propagation over the optical fiber(s) **147** included in the interconnect **145**, which are optically connected to multiple core fibers **137** within the catheter tubing. Herein, the optical receiver **184** is configured to: (i) receive returned optical signals, namely reflected light signals **150** received from optical fiber-based reflective gratings (sensors) fabricated within each of the core fibers **137** deployed within the catheter **130**, and (ii) translate the reflected light signals **150** into reflection data (from repository **192**), namely data in the form of electrical signals representative of the reflected light signals including wavelength shifts caused by strain. The reflected light signals **150** associated with different spectral widths include reflected light signals **151** provided from sensors positioned in the center core fiber (reference) of the catheter **130** and reflected light signals **152** provided from sensors positioned in the outer core fibers of the catheter **130**, as described below.

(43) As noted above, the shape sensing logic **194** is configured to compare wavelength shifts measured by sensors deployed in each outer core fiber at the same measurement region of the catheter (or same spectral width) to the wavelength shift at the center core fiber positioned along central axis and operating as a neutral axis of bending. From these analytics, the shape sensing logic **190** may determine the shape the core fibers have taken in 3D space and may further determine the current physical state of the catheter **130** in 3D space for rendering on the display **170**.

(44) According to one embodiment of the disclosure, the shape sensing logic **194** may generate a rendering of the current physical state of the catheter **130**, especially the integrated tubing, based on heuristics or run-time analytics. For example, the shape sensing logic **194** may be configured in accordance with machine-learning techniques to access a data store (library) with pre-stored data (e.g., images, etc.) pertaining to different regions of the catheter **130** in which the core fibers **137** experienced similar or identical wavelength shifts. From the pre-stored data, the current physical state of the catheter **130** may be rendered. Alternatively, as another example, the shape sensing logic **194** may be configured to determine, during run-time, changes in the physical state of each region of the catheter **130**, notably the tubing, based on at least (i) resultant wavelength shifts experienced by the core fibers **137** and (ii) the relationship of these wavelength shifts generated by sensors positioned along different outer core fibers at the same cross-sectional region of the catheter **130** to the wavelength shift generated by a sensor of the center core fiber at the same cross-sectional region. It is contemplated that other processes and procedures may be performed to utilize the wavelength shifts as measured by sensors along each of the core fibers **137** to render appropriate changes in the physical state of the catheter **130**.

(45) Referring to FIG. 2, an exemplary embodiment of a structure of a section of the multi-core optical fiber included within the stylet **120** of FIG. 1A is shown in accordance with some embodiments. The multi-core optical fiber section **200** of the multi-core optical fiber **135** depicts certain core fibers **137.sub.1-137.sub.M** ($M \geq 2$, $M=4$ as shown, see FIG. 3A) along with the spatial relationship between sensors (e.g., reflective gratings) **210.sub.11-210.sub.NM** ($N \geq 2$; $M \geq 2$) present within the core fibers **137.sub.1-137.sub.M**, respectively. As noted above, the core fibers **137.sub.1-137.sub.M** may be collectively referred to as “the core fibers **137**.”

(46) As shown, the section **200** is subdivided into a plurality of cross-sectional regions **220.sub.1-220.sub.N**, where each cross-sectional region **220.sub.1-220.sub.N** corresponds to reflective gratings **210.sub.11-210.sub.14 . . . 210.sub.N1-210.sub.N4**. Some or all of the cross-sectional regions **220.sub.1 . . . 220.sub.N** may be static (e.g., prescribed length) or may be dynamic (e.g., vary in size among the regions **220.sub.1 . . . 220.sub.N**). A first core fiber **137.sub.1** is positioned substantially along a center (neutral) axis **230** while core fiber **137.sub.2** may be oriented within the cladding of the multi-core optical fiber **135**, from a cross-sectional, front-facing perspective, to be

position on “top” the first core fiber **137.sub.1**. In this deployment, the core fibers **137.sub.3** and **137.sub.4** may be positioned “bottom left” and “bottom right” of the first core fiber **137.sub.1**. As examples, FIGS. **3A-4B** provides illustrations of such.

(47) Referencing the first core fiber **137.sub.1** as an illustrative example, when the stylet **120** is operative, each of the reflective gratings **210.sub.1-210.sub.N** reflects light for a different spectral width. As shown, each of the gratings **210.sub.1i-210.sub.Ni** ($1 \leq i \leq M$) is associated with a different, specific spectral width, which would be represented by different center frequencies of $f.sub.1 \dots f.sub.N$, where neighboring spectral widths reflected by neighboring gratings are non-overlapping according to one embodiment of the disclosure.

(48) Herein, positioned in different core fibers **137.sub.2-137.sub.3** but along at the same cross-sectional regions **220-220.sub.N** of the multi-core optical fiber **135**, the gratings **210.sub.12-210.sub.N2** and **210.sub.13-210.sub.N3** are configured to reflect incoming light at same (or substantially similar) center frequency. As a result, the reflected light returns information that allows for a determination of the physical state of the optical fibers **137** (and the stylet **120**) based on wavelength shifts measured from the returned, reflected light. In particular, strain (e.g., compression or tension) applied to the multi-core optical fiber **135** (e.g., at least core fibers **137.sub.2-137.sub.3**) results in wavelength shifts associated with the returned, reflected light. Based on different locations, the core fibers **137.sub.1-137.sub.4** experience different types and degree of strain based on angular path changes as the stylet **120** advances in the patient.

(49) For example, with respect to the multi-core optical fiber section **200** of FIG. **2**, in response to angular (e.g., radial) movement of the stylet **120** is in the left-veering direction, the fourth core fiber **137.sub.4** (see FIG. **3A**) of the multi-core optical fiber **135** with the shortest radius during movement (e.g., core fiber closest to a direction of angular change) would exhibit compression (e.g., forces to shorten length). At the same time, the third core fiber **137.sub.3** with the longest radius during movement (e.g., core fiber furthest from the direction of angular change) would exhibit tension (e.g., forces to increase length). As these forces are different and unequal, the reflected light from reflective gratings **210.sub.N2** and **210.sub.N3** associated with the core fiber **137.sub.2** and **137.sub.3** will exhibit different changes in wavelength. The differences in wavelength shift of the reflected light signals **150** can be used to extrapolate the physical configuration of the stylet **120** by determining the degrees of wavelength change caused by compression/tension for each of the periphery fibers (e.g., the second core fiber **137.sub.2** and the third core fiber **137.sub.3**) in comparison to the wavelength of the reference core fiber (e.g., first core fiber **137.sub.1**) located along the neutral axis **230** of the multi-core optical fiber **135**. These degrees of wavelength change may be used to extrapolate the physical state of the stylet **120**. The reflected light signals **150** are reflected back to the console **110** via individual paths over a particular core fiber **137.sub.1-137.sub.M**.

(50) Referring to FIG. **3A**, a first exemplary embodiment of the stylet of FIG. **1A** supporting both an optical and electrical signaling is shown in accordance with some embodiments. Herein, the stylet **120** features a centrally located multi-core optical fiber **135**, which includes a cladding **300** and a plurality of core fibers **137.sub.1-137.sub.M** ($M \geq 2$; $M=4$) residing within a corresponding plurality of lumens **320.sub.1-320.sub.M**. While the multi-core optical fiber **135** is illustrated within four (4) core fibers **137.sub.1-137.sub.4**, a greater number of core fibers **137.sub.1-137.sub.M** ($M > 4$) may be deployed to provide a more detailed three-dimensional sensing of the physical state (e.g., shape, etc.) of the multi-core optical fiber **135** and the stylet **120** deploying the optical fiber **135**.

(51) For this embodiment of the disclosure, the multi-core optical fiber **135** is encapsulated within a concentric braided tubing **310** positioned over a low coefficient of friction layer **335**. The braided tubing **310** may feature a “mesh” construction, in which the spacing between the intersecting conductive elements is selected based on the degree of rigidity desired for the stylet **120**, as a greater spacing may provide a lesser rigidity, and thereby, a more pliable stylet **120**.

(52) According to this embodiment of the disclosure, as shown in FIGS. 3A-3B, the core fibers 137.sub.1-137.sub.4 include (i) a central core fiber 137.sub.1 and (ii) a plurality of periphery core fibers 137.sub.2-137.sub.4, which are maintained within lumens 320.sub.1-320.sub.4 formed in the cladding 300. According to one embodiment of the disclosure, one or more of the lumen 320.sub.1-320.sub.4 may be configured with a diameter sized to be greater than the diameter of the core fibers 137.sub.1-137.sub.4. By avoiding a majority of the surface area of the core fibers 137.sub.1-137.sub.4 from being in direct physical contact with a wall surface of the lumens 320.sub.1-320.sub.4, the wavelength changes to the incident light are caused by angular deviations in the multi-core optical fiber 135 thereby reducing influence of compression and tension forces being applied to the walls of the lumens 320.sub.1-320.sub.M, not the core fibers 137.sub.1-137.sub.M themselves.

(53) As further shown in FIGS. 3A-3B, the core fibers 137.sub.1-137.sub.4 may include central core fiber 137.sub.1 residing within a first lumen 320.sub.1 formed along the first neutral axis 230 and a plurality of core fibers 137.sub.2-137.sub.4 residing within lumens 320.sub.2-320.sub.4 each formed within different areas of the cladding 300 radiating from the first neutral axis 230. In general, the core fibers 137.sub.2-137.sub.4, exclusive of the central core fiber 137.sub.1, may be positioned at different areas within a cross-sectional area 305 of the cladding 300 to provide sufficient separation to enable three-dimensional sensing of the multi-core optical fiber 135 based on changes in wavelength of incident light propagating through the core fibers 137.sub.2-137.sub.4 and reflected back to the console for analysis.

(54) For example, where the cladding 300 features a circular cross-sectional area 305 as shown in FIG. 3B, the core fibers 137.sub.2-137.sub.4 may be positioned substantially equidistant from each other as measured along a perimeter of the cladding 300, such as at “top” (12 o'clock), “bottom-left” (8 o'clock) and “bottom-right” (4 o'clock) locations as shown. Hence, in general terms, the core fibers 137.sub.2-137.sub.4 may be positioned within different segments of the cross-sectional area 305. Where the cross-sectional area 305 of the cladding 300 has a distal tip 330 and features a polygon cross-sectional shape (e.g., triangular, square, rectangular, pentagon, hexagon, octagon, etc.), the central core fiber 137.sub.1 may be located at or near a center of the polygon shape, while the remaining core fibers 137.sub.2-137.sub.M may be located proximate to angles between intersecting sides of the polygon shape.

(55) Referring still to FIGS. 3A-3B, operating as the conductive medium for the stylet 120, the braided tubing 310 provides mechanical integrity to the multi-core optical fiber 135 and operates as a conductive pathway for electrical signals. For example, the braided tubing 310 may be exposed to a distal tip of the stylet 120. The cladding 300 and the braided tubing 310, which is positioned concentrically surrounding a circumference of the cladding 300, are contained within the same insulating layer 350. The insulating layer 350 may be a sheath or conduit made of protective, insulating (e.g., non-conductive) material that encapsulates both for the cladding 300 and the braided tubing 310, as shown.

(56) Referring to FIG. 4A, a second exemplary embodiment of the stylet of FIG. 1B is shown in accordance with some embodiments. Referring now to FIG. 4A, a second exemplary embodiment of the stylet 120 of FIG. 1B supporting both an optical and electrical signaling is shown. Herein, the stylet 120 features the multi-core optical fiber 135 described above and shown in FIG. 3A, which includes the cladding 300 and the first plurality of core fibers 137.sub.1-137.sub.M ($M \geq 3$; $M=4$ for embodiment) residing within the corresponding plurality of lumens 320.sub.1-320.sub.M. For this embodiment of the disclosure, the multi-core optical fiber 135 includes the central core fiber 137.sub.1 residing within the first lumen 320.sub.1 formed along the first neutral axis 230 and the second plurality of core fibers 137.sub.2-137.sub.4 residing within corresponding lumens 320.sub.2-320.sub.4 positioned in different segments within the cross-sectional area 305 of the cladding 300. Herein, the multi-core optical fiber 135 is encapsulated within a conductive tubing 400. The conductive tubing 400 may feature a “hollow” conductive cylindrical member

concentrically encapsulating the multi-core optical fiber **135**.

(57) Referring to FIGS. **4A-4B**, operating as a conductive medium for the stylet **120** in the transfer of electrical signals (e.g., ECG signals) to the console, the conductive tubing **400** may be exposed up to a tip **410** of the stylet **120**. For this embodiment of the disclosure, a conductive epoxy **420** (e.g., metal-based epoxy such as a silver epoxy) may be affixed to the tip **410** and similarly joined with a termination/connection point created at a proximal end **430** of the stylet **120**. The cladding **300** and the conductive tubing **400**, which is positioned concentrically surrounding a circumference of the cladding **300**, are contained within the same insulating layer **440**. The insulating layer **440** may be a protective conduit encapsulating both for the cladding **300** and the conductive tubing **400**, as shown.

(58) Referring to FIG. **5A**, an elevation view of a first illustrative embodiment of a catheter including integrated tubing, a diametrically disposed septum, and micro-lumens formed within the tubing and septum is shown in accordance with some embodiments. Herein, the catheter **130** includes integrated tubing, the diametrically disposed septum **510**, and the plurality of micro-lumens **530.sub.1-530.sub.4** which, for this embodiment, are fabricated to reside within the wall **500** of the integrated tubing of the catheter **130** and within the septum **510**. In particular, the septum **510** separates a single lumen, formed by the inner surface **505** of the wall **500** of the catheter **130**, into multiple lumen, namely two lumens **540** and **545** as shown. Herein, the first lumen **540** is formed between a first arc-shaped portion **535** of the inner surface **505** of the wall **500** forming the catheter **130** and a first outer surface **555** of the septum **510** extending longitudinally within the catheter **130**. The second lumen **545** is formed between a second arc-shaped portion **565** of the inner surface **505** of the wall **500** forming the catheter **130** and a second outer surfaces **560** of the septum **510**.

(59) According to one embodiment of the disclosure, the two lumens **540** and **545** have approximately the same volume. However, the septum **510** need not separate the tubing into two equal lumens. For example, instead of the septum **510** extending vertically (12 o'clock to 6 o'clock) from a front-facing, cross-sectional perspective of the tubing, the septum **510** could extend horizontally (3 o'clock to 9 o'clock), diagonally (1 o'clock to 7 o'clock; 10 o'clock to 4 o'clock) or angularly (2 o'clock to 10 o'clock). In the later configuration, each of the lumens **540** and **545** of the catheter **130** would have a different volume.

(60) With respect to the plurality of micro-lumens **530.sub.1-530.sub.4**, the first micro-lumen **530.sub.1** is fabricated within the septum **510** at or near the cross-sectional center **525** of the integrated tubing. For this embodiment, three micro-lumens **530.sub.2-530.sub.4** are fabricated to reside within the wall **500** of the catheter **130**. In particular, a second micro-lumen **530.sub.2** is fabricated within the wall **500** of the catheter **130**, namely between the inner surface **505** and outer surface **507** of the first arc-shaped portion **535** of the wall **500**. Similarly, the third micro-lumen **530.sub.3** is also fabricated within the wall **500** of the catheter **130**, namely between the inner and outer surfaces **505/507** of the second arc-shaped portion **555** of the wall **500**. The fourth micro-lumen **530.sub.4** is also fabricated within the inner and outer surfaces **505/507** of the wall **500** that are aligned with the septum **510**.

(61) According to one embodiment of the disclosure, as shown in FIG. **5A**, the micro-lumens **530.sub.2-530.sub.4** are positioned in accordance with a “top-left” (10 o'clock), “top-right” (2 o'clock) and “bottom” (6 o'clock) layout from a front-facing, cross-sectional perspective. Of course, the micro-lumens **530.sub.2-530.sub.4** may be positioned differently, provided that the micro-lumens **530.sub.2-530.sub.4** are spatially separated along the circumference **520** of the catheter **130** to ensure a more robust collection of reflected light signals from the outer core fibers **570.sub.2-570.sub.4** when installed. For example, two or more of micro-lumens (e.g., micro-lumens **530.sub.2** and **530.sub.4**) may be positioned at different quadrants along the circumference **520** of the catheter wall **500**.

(62) Referring to FIG. **5B**, a perspective view of the first illustrative embodiment of the catheter of

FIG. 5A including core fibers installed within the micro-lumens is shown in accordance with some embodiments. According to one embodiment of the disclosure, the second plurality of micro-lumens **530.sub.2-530.sub.4** are sized to retain corresponding outer core fibers **570.sub.2-570.sub.4**, where the diameter of each of the second plurality of micro-lumens **530.sub.2-530.sub.4** may be sized just larger than the diameters of the outer core fibers **570.sub.2-570.sub.4**. The size differences between a diameter of a single core fiber and a diameter of any of the micro-lumen **530.sub.1-530.sub.4** may range between 0.001 micrometers (μm) and 1000 μm , for example. As a result, the cross-sectional areas of the outer core fibers **570.sub.2-570.sub.4** would be less than the cross-sectional areas of the corresponding micro-lumens **530.sub.2-530.sub.4**. A “larger” micro-lumen (e.g., micro-lumen **530.sub.2**) may better isolate external strain being applied to the outer core fiber **570.sub.2** from strain directly applied to the catheter **130** itself. Similarly, the first micro-lumen **530.sub.1** may be sized to retain the center core fiber **570.sub.1**, where the diameter of the first micro-lumen **530.sub.1** may be sized just larger than the diameter of the center core fiber **570.sub.1**.

(63) As an alternative embodiment of the disclosure, one or more of the micro-lumens **530.sub.1-530.sub.4** may be sized with a diameter that exceeds the diameter of the corresponding one or more core fibers **570.sub.1-570.sub.4**. However, at least one of the micro-lumens **530.sub.1-530.sub.4** is sized to fixedly retain their corresponding core fiber (e.g., core fiber retained with no spacing between its lateral surface and the interior wall surface of its corresponding micro-lumen). As yet another alternative embodiment of the disclosure, all the micro-lumens **530.sub.1-530.sub.4** are sized with a diameter to fixedly retain the core fibers **570.sub.1-570.sub.4**.

(64) Referring to FIGS. **6A-6B**, flowcharts of methods of operations conducted by the medical instrument monitoring system of FIGS. **1A-1B** to achieve optic 3D shape sensing are shown in accordance with some embodiments. Herein, the catheter includes at least one septum spanning across a diameter of the tubing wall and continuing longitudinally to subdivide the tubing wall. The medial portion of the septum is fabricated with a first micro-lumen, where the first micro-lumen is coaxial with the central axis of the catheter tubing. The first micro-lumen is configured to retain a center core fiber. Two or more micro-lumen, other than the first micro-lumen, are positioned at different locations circumferentially spaced along the wall of the catheter tubing. For example, two or more of the second plurality of micro-lumens may be positioned at different quadrants along the circumference of the catheter wall.

(65) Furthermore, each core fiber includes a plurality of sensors spatially distributed along its length between at least the proximal and distal ends of the catheter tubing. This array of sensors is distributed to position sensors at different regions of the core fiber to enable distributed measurements of strain throughout the entire length or a selected portion of the catheter tubing. These distributed measurements may be conveyed through reflected light of different spectral widths (e.g., specific wavelength or specific wavelength ranges) that undergoes certain wavelength shifts based on the type and degree of strain.

(66) According to one embodiment of the disclosure, as shown in FIG. **6A**, for each core fiber, broadband incident light is supplied to propagate through a particular core fiber (block **600**). Unless discharged, upon the incident light reaching a sensor of a distributed array of sensors measuring strain on a particular core fiber, light of a prescribed spectral width associated with the first sensor is to be reflected back to an optical receiver within a console (blocks **605-610**). Herein, the sensor alters characteristics of the reflected light signal to identify the type and degree of strain on the particular core fiber as measured by the first sensor (blocks **615-620**). According to one embodiment of the disclosure, the alteration in characteristics of the reflected light signal may signify a change (shift) in the wavelength of the reflected light signal from the wavelength of the incident light signal associated with the prescribed spectral width. The sensor returns the reflected light signal over the core fiber and the remaining spectrum of the incident light continues propagation through the core fiber toward a distal end of the catheter tubing (blocks **625-630**). The

remaining spectrum of the incident light may encounter other sensors of the distributed array of sensors, where each of these sensors would operate as set forth in blocks **605-630** until the last sensor of the distributed array of sensors returns the reflected light signal associated with its assigned spectral width and the remaining spectrum is discharged as illumination.

(67) Referring now to FIG. **6B**, during operation, multiple reflected light signals are returned to the console from each of the plurality of core fibers residing within the corresponding plurality of micro-lumens formed within a catheter, such as the catheter of FIG. **1B**. In particular, the optical receiver receives reflected light signals from the distributed arrays of sensors located on the center core fiber and the outer core fibers and translates the reflected light signals into reflection data, namely electrical signals representative of the reflected light signals including wavelength shifts caused by strain (blocks **650-655**). The reflection data classification logic is configured to identify which core fibers pertain to which reflection data and segregate reflection data provided from reflected light signals pertaining to a particular measurement region (or similar spectral width) into analysis groups (block **660-665**).

(68) Each analysis group of reflection data is provided to shape sensing logic for analytics (block **670**). Herein, the shape sensing logic compares wavelength shifts at each outer core fiber with the wavelength shift at the center core fiber positioned along central axis and operating as a neutral axis of bending (block **675**). From this analytics, on all analytic groups (e.g., reflected light signals from sensors in all or most of the core fibers), the shape sensing logic may determine the shape the core fibers have taken in three-dimensional space, from which the shape sensing logic can determine the current physical state of the catheter in three-dimension space (blocks **680-685**).

(69) Referring to FIG. **7**, an exemplary embodiment of the medical instrument monitoring system of FIG. **1B** during operation and insertion of the catheter into a patient are shown in accordance with some embodiments. Herein, the catheter **130** generally includes the integrated tubing of the catheter **130** with a proximal portion **720** that generally remains exterior to the patient **700** and a distal portion **730** that generally resides within the patient vasculature after placement is complete. The (integrated) catheter tubing of the catheter **130** may be advanced to a desired position within the patient vasculature such as a distal end (or tip) **735** of the catheter tubing of the catheter **130** is proximate the patient's heart, such as in the lower one-third ($\frac{1}{3}$) portion of the Superior Vena Cava ("SVC") for example. In some embodiments, various instruments may be disposed at the distal end **735** of the catheter **130** to measure pressure of blood in a certain heart chamber and in the blood vessels, view an interior of blood vessels, or the like. In alternative embodiments, such as those that utilize the stylet assembly of FIG. **1A** and the catheter **195**, such instruments may be disposed at a distal end of the stylet **120**.

(70) During advancement through a patient vasculature, the catheter tubing of the catheter **130** receives broadband incident light **155** from the console **110** via optical fiber(s) **147** within the interconnect **145**, where the incident light **155** propagates along the core fibers **137** of the multi-core optical fiber **135** within the catheter tubing of the catheter **130**. According to one embodiment of the disclosure, the connector **146** of the interconnect **145** terminating the optical fiber(s) **147** may be coupled to the optical-based catheter connector **144**, which may be configured to terminate the core fibers **137** deployed within the catheter **130**. Such coupling optically connects the core fibers **137** of the catheter **130** with the optical fiber(s) **147** within the interconnect **145**. The optical connectivity is needed to propagate the incident light **155** to the core fibers **137** and return the reflected light signals **150** to the optical logic **180** within the console **110** over the interconnect **145**. As described below in detail, the physical state of the catheter **130** may be ascertained based on analytics of the wavelength shifts of the reflected light signals **150**.

(71) Referring to FIG. **8A**, a perspective illustration of 3D catheter images within a plane of a 3D Cartesian coordinate system is shown in accordance with some embodiments. Specifically, the plane **800** illustrates a Cartesian coordinate system for a three-dimensional space consisting of an ordered triplet of lines (axes) commonly referred to the x-axis, y-axis and z-axis, where a point

within the plane **800** is represented by a numerical value along each axis as (x, y, z), often called “coordinates.” The axes originate at a common point, which is often referred to as the “origin,” having coordinates within the plane **800** of (0, 0, 0)

(72) As seen in FIG. **8A**, a plurality of 3D catheter images being 3D representations **802.sub.1-802.sub.i** of the physical state of a stylet, i.e., the stylet **120**, advancing through the vasculature of a patient's body. The 3D representations **802.sub.1-802.sub.i** are computed from characteristics of reflected light signals by shape sensing logic as discussed above. However, what is notably missing from the computed characteristics of the reflected light signals **150** and the ensuing 3D representations **802.sub.1-802.sub.i** is an indication of an orientation of the catheter relative to the patient's body.

(73) Having knowledge of the orientation of the catheter relative to the patient's body is integral in generating and rendering a 2D display of the shape of the stylet (and/or catheter), as most, if not all, displays utilized in catheter tracking systems are 2D displays, e.g., such as the display **170**. Thus, without an indication of the orientation of the catheter relative to the patient's body, a 2D display cannot be rendered that properly depicts the stylet on a display, e.g., the display **170**, that is anatomically correct relative to the reference frame of the patient's body.

(74) To further explain this problem, FIG. **8A** illustrates the plurality of 3D representations **802.sub.1-802.sub.i** of the physical state of a stylet advancing through the vasculature of a patient having various orientations within the plane **800**. Each of the 3D representations **802.sub.1-802.sub.i** may be in accordance with the characteristics of received reflected light signals, which merely indicate a shape such as a curve of the stylet without an indication as to a direction of the curve relative to a frame of reference. Thus, in order to generate and render a 2D display of the catheter relative to the reference frame of the patient's body, an orientation of the stylet must be determined in accordance with the reference frame of the patient's body.

(75) Referring to now FIG. **8B**, a perspective illustration of 3D catheter images within the plane **810** of a 3D Cartesian coordinate system is shown in accordance with some embodiments. FIG. **8B** illustrates a first step according to one embodiment of determining the orientation of the 3D representations **802.sub.1-802.sub.i** of FIG. **8A**. As discussed in further detail with respect to the embodiment of FIGS. **9A-9B**, the percutaneous insertion site (“insertion site”) of the stylet (and catheter, collectively referred to as the “stylet”) may be known (e.g., on a patient's arm distal the elbow joint) and the initial direction that the stylet advances relative to the patient's body is known (e.g., toward the clavicle). As a result, a reference frame for the patient's body may be established by logic of the console in accordance with a coordinate system (such as an abstract x,y coordinate system) in relation to the patient. Therefore, the initial direction of the advancement of the stylet may be established as the positive direction along the y-axis with respect to the x,y coordinate system of the reference frame of the patient. As an illustrative example, FIG. **10** depicts of an exemplary embodiment of the medical instrument monitoring system illustrating an established reference frame **1020** of the patient, which is mirrored by an established reference frame **1022** of the display **170**, with each frame of reference utilizing an x,y coordinate system.

(76) Referring again to FIG. **8B**, by constraining the initial direction of the advancement of the stylet to be in the positive direction along the y-axis, the reference frame establishment logic **196** may record and establish a correlation between initial reflected light received and the positive direction of the y-axis. Such a correlation may be stored by the reference frame establishment logic **196** and used to in defining the direction of the advancement of the stylet as discussed below. The term “initial reflected light” may refer to the portion of the received reflected light that corresponds to an indication that the stylet is traveling in a straight or substantially path. For example, with reference again to FIG. **10**, the initial reflected light may correspond to the reflected light received while the stylet is advancing through segment **1002** of the patient **1000**. Thus, a 2D depiction of the catheter advancing through the vasculature of the patient from the insertion site toward the patient's clavicle may be displayed on a 2D display based on the received reflected light and the constraint

of the initial advancement of the stylet to be in the positive direction along the y-axis.

(77) Continuing the example, as the stylet advances toward the clavicle, the stylet eventually begins to curve toward the patient's rib cage (i.e., toward the center of the patient's body). FIG. 8B illustrates the multiple scenarios that exist when the direction of the initial advancement of the stylet is known but the direction the stylet is curving is known. In other words, even though the direction of the initial advancement of the stylet is known, once the stylet begins to curve within the body, the received reflected light signals merely provide an indication of the angle of the curve, not direction of the curve relative to the reference frame of the patient.

(78) Referring again to FIG. 10, the detection of the curve may occur as the stylet advances through the segment 1004. Referring now back to FIG. 8B, upon detection of the curvature of the stylet by the shape sensing logic 194, the frame reference establishment logic 196 establishes a correlation between the received reflected light and the positive direction along the x-axis. This is done based on knowledge that the stylet is constrained anatomically to curve toward the center of the patient's chest, which is in the positive direction along the x-axis in accordance with the established reference frame of the patient.

(79) Thus, following detection of the curvature of the stylet, the frame reference establishment logic 196 establishes an orientation of the stylet, and thus, the catheter. By establishing a correlation between the initial reflected light and the positive direction along the y-axis and a correlation between reflected light corresponding to a curvature of the stylet, the frame reference establishment logic 196 can generate and render a 2D display illustrating a representation of the shape of the catheter during advancement through the vasculature of the patient having a proper anatomical orientation based on the reference frame of the patient's body. As the stylet continues its advancement, the shape sensing logic 194 in conjunction with the reference frame establishment logic 196 may generate a 2D display of the catheter by comparing the received reflected light with the orientation of the stylet defined by (i) the initial reflected light signals correlated with the positive direction along the y-axis of the reference frame of the patient, and (ii) the reflected light signals corresponding to the curve in the advancement correlated with the positive direction along the x-axis of the reference frame of the patient.

(80) Referring now to FIGS. 9A-9B, a flowchart of a first embodiment of operations performed by logic of the medical instrument monitoring system of FIG. 1A is shown in accordance with some embodiments. Each block illustrated in FIGS. 9A-9B represents an operation performed in the method 900 of detecting an orientation of a catheter disposed within a patient's vasculature based on received reflect light signals. It is assumed that prior to the beginning of the method 900, a catheter having a stylet disposed therein has been inserted into the patient's vasculature at a known percutaneous insertion site (e.g., at a location on the patient's arm distal the elbow joint). Specifically, it is assumed that the stylet includes a multi-core optical fiber and is coupled to a console configured to provide broadband incident light to the stylet, such as the console 110 and the stylet 120 of FIG. 1A.

(81) Thus, the method 900 begins when broadband incident light is received by and propagates along the multi-core optical fiber of the stylet. The broadband incident light is reflected by one or more gratings (sensors) of the multi-core optical fiber, such that the reflected light signals may include wavelength shifts that correlate with strain experienced by each sensor, as discussed above. The console receives the reflected optical signals (reflected light signals) from the gratings fabricated within each core fiber of the multi-core optical fiber deployed within the stylet (block 902).

(82) The reflect light signals are then translated into reflection data being electrical signals representing wavelength shifts in each core fiber caused by strain on the core fiber during advancement of the stylet through the patient's vasculature (block 904). Analytics are then performed to compare the wavelength shifts of periphery core fibers to a center core fiber operating as a neutral axis of bending to determine the shape of core fibers (i.e., that which the stylet and the

catheter) have taken in 3D space (block **906**).

(83) Following performance of analytics to determine a 3D shape of the catheter, a 2D display is generated and rendered to display a 2D representation of the shape of the catheter as the catheter advances through the patient's vasculature using an x,y coordinate system (block **908**). The 2D display is rendered such that the initial advancement of the catheter is illustrated as being in the positive direction along the y-axis. As noted above, the x,y coordinate system of the display, such as the display **170** of FIG. **1A**, corresponds to a reference frame that matches a reference frame of the patient's body. Thus, the 2D display of the catheter during its initial advancement matches the anatomical positioning of the catheter's initial advancement within the patient's body.

(84) Referring now to FIG. **9B**, reflected optical signals (light signals) continue to be received from the multi-core optical fiber providing a continued indication of the shape of the stylet, which corresponds to the advancement of the catheter within the patient's vasculature. Analytics are performed on the received reflected light signals in order to continue determining the shape that the stylet, and catheter, has taken in 3D shape in order to continue monitoring its advancement (block **910**). As reflected light signals are continually received and the 3D shape of the catheter is determined during an initial advancement of the catheter, a determination is made as to whether the reflection data derived from the reflected light signals indicates that the stylet has curved during advancement (block **912**). When a result of the determination is that the stylet has not curved (no at block **912**), generation and display of the 2D display is continued in which the advancement of the catheter is illustrated in the positive direction along the y-axis of the reference frame of the display (e.g., the display **170**) (block **914**).

(85) When a result of the determination is that the stylet has begun to curve (yes at block **912**), an orientation of the stylet is established using the reference frame of the 2D display to match a reference frame of the patient's body (block **916**). Specifically, the direction of the curve indicated by the reflected light signals is set to correlate with the positive direction along the x-axis of the 2D display. The orientation of the stylet with respect to the reference frame of the 2D display is established based on the direction of the initial advancement of the stylet being set to correlate with the positive direction along the y-axis and the direction of the curve being set to correlate with the positive direction along the x-axis. Thus, the logic of the console has established the orientation of the stylet (and catheter) such that display of the advancement of the catheter on the 2D display is illustrated in an anatomically proper manner with respect to the reference frame of the patient's body. The established orientation may be stored as the correlation of the reflected light signals (e.g., specific wavelength shifts) with a particular direction along a particular axis in the console, e.g., along with the reflect data **192** of FIGS. **1A-1B** or, alternatively, in a separate datastore of console **110** not shown.

(86) The 2D display is continually generated and rendered in order to illustrate advancement of the catheter based on received optical signals (light signals) utilizing the established orientation of the stylet (and catheter) with respect to the reference frame of the 2D display that matches that of the patient's body (block **918**).

(87) Referring to FIG. **10**, a second exemplary embodiment of the medical instrument monitoring system of FIG. **1A** during operation and insertion of a catheter including a stylet into a patient is shown in accordance with some embodiments. As noted above, FIG. **10** illustrates a series of theoretical segments of the stylet **120** (segments **1002**, **1004**, **1006**, **1008**), where each segment may indicate a transition from a first shape of the stylet **120** (and the catheter **195**) to a second shape. As further illustrated, the catheter **195** generally includes a proximal portion **1011** that generally remains exterior to the patient **1000** and a distal portion **1012** that generally resides within the patient vasculature after placement is complete. The stylet **120** is employed to assist in the positioning of the distal tip **1014** of the catheter **195** in a desired position within the patient vasculature. In one embodiment, the desired position for the catheter distal tip **1014** is proximate the patient's heart, such as in the lower one-third ($\frac{1}{3}$ rd) portion of the Superior Vena Cava ("SVC")

for this embodiment. Of course, the stylet **120** can be employed to place the catheter distal tip **1014** in other locations.

(88) During advancement, the stylet **120** receives broadband light **155** from the console **110** via interconnect **145**, which includes the connector **146** for coupling to the console connector **144** for the stylet assembly **118**. The reflected light **150** from sensors (reflective gratings) within each core fiber of the multi-core optical fiber **137** are returned from the stylet **120** over the interconnect **145** for processing by the console **110**. The physical state of the stylet **120** may be ascertained based on analytics of the wavelength shifts of the reflected light **150**. For example, the strain caused through bending of the stylet **120**, and hence angular modification of each core fiber, causes different degrees of deformation. The different degrees of deformation alters the shape of the sensors (reflective grating) positioned on the core fiber, which may cause variations (shifts) in the wavelength of the reflected light from the sensors positioned on each core fiber within the multi-core optical fiber **137**, as shown in FIG. 2. From this wavelength shifting, the shape sensing analytic logic **194** within the console **110** may determine the physical state of the stylet **120** (e.g., shape, etc.), and as the catheter **195** mirrors the stylet **120**, the physical state of the catheter **195** is also determined.

(89) As discussed briefly above, FIG. 10 illustrates two reference frames: a reference frame of the patient's body comprising the coordinate system **1016** relative to the patient's body; and a reference frame of the display **170** that mirrors the orientation of the reference frame of the patient's body and comprises the coordinate system **1018** relative to the boundaries of the display **170**. Thus, referring to the embodiment of method **900** of FIGS. 9A-9B, the orientation of the reference frame of the patient's body (including the coordinate system **1016**) is equivalent to the reference frame of the display **170** (including the coordinate system **1018**). Thus, by determining an orientation of the stylet **120**, and hence the catheter **195**, the logic of the console **110** may generate a render a depiction of a 2D representation of the shape of the catheter **195** as it advances within the vasculature of the patient's body in an anatomically proper manner.

(90) Referring to FIGS. 11A-11B, a flowchart of a second embodiment of operations performed by logic of the medical instrument monitoring system of FIG. 1A is shown in accordance with some embodiments. Each block illustrated in FIGS. 11A-11B represents an operation performed in the method **1100** of detecting an orientation of a catheter disposed within a patient's vasculature based on received reflect light signals. The assumptions discussed above with respect to the method **900** apply equally with respect to the method **1100**.

(91) Thus, the method **1100** begins when broadband incident light is received by and propagates along the multi-core optical fiber of the stylet. The broadband incident light is reflected by one or more gratings (sensors) of the multi-core optical fiber, such that the reflected light signals may include wavelength shifts that correlate with strain experienced by each sensor, as discussed above. The console receives the reflected optical signals (reflected light signals) from the gratings fabricated within each core fiber of the multi-core optical fiber deployed within the stylet (block **1102**).

(92) The reflect light signals are then translated into reflection data being electrical signals representing wavelength shifts in each core fiber caused by strain on the core fiber during advancement of the stylet through the patient's vasculature (block **1104**). Analytics are then performed to compare the wavelength shifts of periphery core fibers to a center core fiber operating as a neutral axis of bending to determine the shape of core fibers (i.e., that which the stylet and the catheter) have taken in 3D space (block **1106**).

(93) Following performance of analytics to determine a 3D shape of the catheter, a 2D display is generated and rendered to display a 2D representation of the shape of the catheter as the catheter advances through the patient's vasculature using an x,y coordinate system relative to a reference frame of the patient as discussed above with respect to FIGS. 9A-9B (block **1108**). The 2D display is rendered such that the direction of the initial advancement of the catheter is illustrated as being in

the positive direction along the y-axis as discussed above with respect to FIGS. 9A-9B.

(94) Referring now to FIG. 11B, reflected optical signals (light signals) continue to be received from the multi-core optical fiber providing a continued indication of the shape of the stylet, which corresponds to the advancement of the catheter within the patient's vasculature. Heuristics or run-time analytics are performed on the reflected light signals to correlate the 3D shape of the stylet to a plurality of pre-stored images (block 1110). In one such embodiment, a machine-learning model is provided with the current 3D shape of the stylet, which is processed to affect a correlation of the current 3D shape with pre-stored 3D images. The results of the processing of the machine-learning model indicate a correlation value between the current 3D shape and one or more of the pre-stored images, which each have a known orientation relative to the patient's body. Therefore, when a correlation value exceeds a threshold value, the current 3D shape is determined to have the known orientation of the corresponding pre-stored image. As would be understood by one of ordinary skill in the art, the machine-learning model would be pre-trained using the pre-stored images of 3D shapes of stylets/catheters each having a known orientation relative to the reference frame of the patient. Further, the machine-learning algorithm utilized may be any known machine-learning algorithm, especially, but not limited or restricted to supervised learning algorithms such as logistic regression and/or neural networks.

(95) Thus, based on results of the heuristics or run-time analytics, an orientation of the stylet is established relative to the reference frame of the patient's body, which is mirrored by the reference frame of the 2D (block 1112). Thus, the console has established the orientation of the stylet (and catheter) such that display of the advancement of the catheter on the 2D display is illustrated in an anatomically proper manner with respect to the reference frame of the patient's body. The 2D display is continually generated and rendered in order to illustrate advancement of the catheter based on received optical signals (light signals) utilizing the established orientation of the stylet (and catheter) with respect to the reference frame of the 2D display that matches that of the patient's body (block 1114).

(96) While some particular embodiments have been disclosed herein, and while the particular embodiments have been disclosed in some detail, it is not the intention for the particular embodiments to limit the scope of the concepts provided herein. Additional adaptations and/or modifications can appear to those of ordinary skill in the art, and, in broader aspects, these adaptations and/or modifications are encompassed as well. Accordingly, departures may be made from the particular embodiments disclosed herein without departing from the scope of the concepts provided herein.

Claims

1. A non-transitory computer-readable medium having stored thereon logic that, when executed by one or more processors of a medical instrument monitoring system, causes operations including: providing a broadband incident light signal to each of a plurality of reflective gratings distributed along a length of each of a plurality of core fibers of a multi-core optical fiber, the plurality of core fibers being spatially distributed to experience different degrees of strain; receiving reflected light signals of different spectral widths of the broadband incident light signal by each of the plurality of reflective gratings; processing the reflected light signals received from each of the plurality of reflective gratings associated with the plurality of core fibers to determine: (i) a physical state of the multi-core optical fiber relating to a medical device including the multi-core optical fiber, and (ii) an orientation of the multi-core optical fiber relative to a reference frame of a body of a patient and a coordinate system thereof, wherein determining the orientation of the multi-core optical fiber relative to the reference frame of the body includes: establishing the reference frame of the body utilizing the coordinate system; and correlating initial reflected light signals to an initial direction of advancement along a first axis of the coordinate system for the multi-core optical fiber based on

the multi-core optical fiber entering the body at a known insertion site; and generating a display illustrating the physical state of the multi-core optical fiber based on at least the orientation determined during processing of the reflected light signals, the coordinate system of the reference frame of the body of the patient oriented with a coordinate system of a reference frame of the display.

2. The non-transitory computer-readable medium of claim 1, wherein the display is a two-dimensional representation of the physical state of the multi-core optical fiber in accordance with the orientation determined during processing of the reflected light signals.
3. The non-transitory computer-readable medium of claim 1, wherein the physical state of the multi-core optical fiber relating to the medical device includes one or more of a length, a shape, or a form as instantly possessed by the multi-core optical fiber.
4. The non-transitory computer-readable medium of claim 1, wherein the different degrees of strain include compression and tension.
5. The non-transitory computer-readable medium of claim 1, wherein the medical device includes an elongated shape and is inserted into a vasculature of the body of the patient.
6. The non-transitory computer-readable medium of claim 5, wherein the medical device is a stylet removably inserted into a lumen of a catheter assembly for placement of a distal tip of the catheter assembly in a superior vena cava of the vasculature.
7. The non-transitory computer-readable medium of claim 1, wherein at least two of the plurality of core fibers experience different types of strain in response to changes in the orientation of the multi-core optical fiber.
8. The non-transitory computer-readable medium of claim 1, wherein each reflective grating of the plurality of reflective gratings alters its reflected light signal by applying a wavelength shift dependent on a strain experienced by the plurality of reflective gratings.
9. The non-transitory computer-readable medium of claim 1, wherein each coordinate system of the reference frame of the body of the patient and the coordinate system of the reference frame of the display is a Cartesian coordinate system.
10. The non-transitory computer-readable medium of claim 9, wherein the coordinate system of the reference frame of the body of the patient is a 3-dimensional ("3D") coordinate system.
11. The non-transitory computer-readable medium of claim 9, wherein the coordinate system of the reference frame of the display is a 2-dimensional ("2D") coordinate system.
12. The non-transitory computer-readable medium of claim 11, wherein generating the display includes generating a 2D display of a catheter as the medical device with shape sensing logic in conjunction with reference frame establishment logic by comparing the reflected light signals with an orientation of a stylet within the catheter defined by: (i) the initial reflected light signals correlated with a positive direction along a y-axis of the reference frame of the body of the patient, and (ii) the reflected light signals corresponding to a curve in an advancement correlated with a positive direction along an x-axis of the reference frame of the patient.
13. The non-transitory computer-readable medium of claim 11, further comprising performing analytics to compare wavelength shifts of periphery core fibers in the multi-core optical fiber to a center core fiber thereof operating as a neutral axis of bending and determine a shape of the periphery core fibers in 3-dimensional ("3D") space.
14. The non-transitory computer-readable medium of claim 1, wherein determining the orientation of the multi-core optical fiber relative to the reference frame of the body includes establishing the initial direction of advancement along the first axis of the coordinate system for the multi-core optical fiber based on the multi-core optical fiber entering the body at the known insertion site.
15. The non-transitory computer-readable medium of claim 1, wherein determining the orientation of the multi-core optical fiber relative to the reference frame of the body includes detecting a curve in advancement along the first axis of the coordinate system for the multi-core optical fiber based on processing the reflected light signals.

16. The non-transitory computer-readable medium of claim 1, wherein determining the orientation of the multi-core optical fiber relative to the reference frame of the body includes correlating the reflected light signals corresponding to a curve in advancement along the first axis of the coordinate system for the multi-core optical fiber with a second direction of advancement along a second axis of the coordinate system.
17. The non-transitory computer-readable medium of claim 16, wherein the orientation of the multi-core optical fiber relative to the reference frame of the body is defined by: (i) the initial reflected light signals correlated with the initial direction of advancement along the first axis of the coordinate system for the multi-core optical fiber based on the multi-core optical fiber entering the body at the known insertion site.
18. The non-transitory computer-readable medium of claim 17, wherein the orientation of the multi-core optical fiber relative to the reference frame of the body is further defined by: (ii) the reflected light signals corresponding to the curve in the advancement correlated with the second direction of advancement along the second axis of the coordinate system.
19. A non-transitory computer-readable medium having stored thereon logic that, when executed by one or more processors of a medical instrument monitoring system, causes operations including: providing a broadband incident light signal to each of a plurality of reflective gratings distributed along a length of each of a plurality of core fibers of a multi-core optical fiber, the plurality of core fibers being spatially distributed to experience different degrees of strain; receiving reflected light signals of different spectral widths of the broadband incident light signal by each of the plurality of reflective gratings; processing the reflected light signals received from each of the plurality of reflective gratings associated with the plurality of core fibers to determine: (i) a physical state of the multi-core optical fiber relating to a medical device including the multi-core optical fiber, and (ii) an orientation of the multi-core optical fiber relative to a reference frame of a body of a patient and a coordinate system thereof, wherein determining the orientation of the multi-core optical fiber relative to the reference frame of the body includes: establishing the reference frame of the body utilizing the coordinate system; establishing an initial direction of advancement along a first axis of the coordinate system for the multi-core optical fiber based on the multi-core optical fiber entering the body at a known insertion site; correlating initial reflected light signals to the initial direction of advancement along the first axis of the coordinate system; detecting a curve in the advancement of the multi-core optical fiber based on processing of the reflected light signals; and correlating the reflected light signals corresponding to the curve in the advancement with a second direction of advancement along a second axis of the coordinate system, wherein the orientation is defined by: (i) the initial reflected light signals correlated with the initial direction of advancement along the first axis of the coordinate system, and (ii) the reflected light signals corresponding to the curve in the advancement correlated with the second direction of advancement along the second axis of the coordinate system; and generating a display illustrating the physical state of the multi-core optical fiber based on at least the orientation determined during processing of the reflected light signals, the coordinate system of the reference frame of the body of the patient oriented with the coordinate system of the reference frame of the display.
20. The non-transitory computer-readable medium of claim 19, wherein the medical device is a stylet removably inserted into a lumen of a catheter assembly for placement of a distal tip of the catheter assembly in a superior vena cava of a vasculature of the patient.
21. The non-transitory computer-readable medium of claim 19, wherein generating the display includes generating a 2D display of a catheter as the medical device with shape sensing logic in conjunction with reference frame establishment logic by comparing the reflected light signals with an orientation of a stylet within the catheter defined by: (i) the initial reflected light signals correlated with a positive direction along a y-axis of the reference frame of the body of the patient, and (ii) the reflected light signals corresponding to the curve in advancement correlated with a positive direction along an x-axis of the reference frame of the patient.

